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1 Introduction

1.1 Objective of this Document

The objective of this document is to describe the Concept of Operations (CONOPS) upon which the development of standardisation material will be prepared for the Operational Excellence Programme (OEP) workstream topic 12.2 to support implementation of the CP1 regulation [Ref 11] on ATM Functionality 6 (AF6) (Initial Trajectory Sharing).

This CONOPS corresponds to deliverable DEL2.1-1 of the topic plan [Ref 1]. An initial draft of this deliverable was presented and submitted to consultation during a public stakeholder workshop¹ September 13th, 2022. This CONOPS will be maintained during the execution of the workstream topic 12.2.

1.2 Background

1.2.1 Initial Trajectory Sharing

SESAR project PJ31 DIGITS led to the very large demonstrator validating the use of ADS-C/EPP and data collection from about 20,000 flights between 2019 and 2020. The results are available at the following website: <https://cordis.europa.eu/project/id/731818/results>.

Deliverable D1.2 details the demonstration of ATM improvements generated by initial trajectory sharing for airspace users. It includes several valuable conclusions and recommendations regarding ATS B2. This led to the development of the CP1 regulation on ATM functionality 6 (CP1 AF6 – Initial Trajectory Sharing).

1.2.2 Regulatory Context

Following the publication of the CP1 regulation [Ref 11], the downlinking and processing of Automatic Dependent Surveillance - Contract (ADS-C) extended projected profile (EPP) as part of ATS B2 is mandated. It is a forward-fitting mandate (from 31/12/2027), applicable to all GAT flying in the SES airspace above FL285. By that date, ANSPs and NM must be able to receive and process downlinked trajectory information from equipped aircraft. ANSP systems must allow controllers to display the route of the downlinked trajectory and warn controllers in cases of discrepancy between downlinked and ground system trajectories.

An important milestone of CP1 AF6 is the industrial target date (or maturity gate) of 31 December 2023, whereby the EC supported by EASA will determine whether AF6 functionalities or sub-functionalities *“have been standardised and that they are ready for implementation”*. Therefore, it is key to support the industrial target date, regarding the needs identified within the SESAR Deployment Programme [Ref 13] for AF6. The OEP Workstream 12.2 will deliver a CONOPS and complementary specifications supporting the recommended implementation of an ADS-C Common Service for CP1 AF6.

1.2.3 Goals and Benefits of an ADS-C Common Service

The demonstration of an ADS-C common service to collect and distribute ADS-C data from flights has been achieved by SESAR PJ38 (ADSCENSIO).

¹ <https://www.eurocontrol.int/event/operational-excellence-programme-ats-b2-workshop-ads-c-common-service-conops>

The SESAR PJ38 deliverables [Ref 2] identified goals and benefits of such an ADS-C common service. These can be summarised in the following terms which highlight the necessity of building a CONOPS for AF6 deployment based on an ADS-C common service:

- **Optimisation and Efficiency**

Optimise the use of air-ground and airborne resources (radio spectrum, ATN VDLm2 network load, number of simultaneous contracts) by avoiding multiple ground users establishing ADS-C contracts to receive the same data.

- **User Scalability**

Increase the number of served ground ADS-C users independently of airborne limitations on the maximum number of ADS-C connections and peers.

- **Services Evolutions and Economies of Scale**

Improved pan-European ADS-C service roll-out of new standard functionalities requiring limited local developments to receive ADS-C data enabling economies of scale.

- **Accelerated Deployment**

Accelerated service roll-out independent of ANSP in-house developments through common initiatives. Immediate and widespread access to ADS-C data for ground users, which is only constrained by air-ground service coverage.

- **Harmonised Common Services**

A common pan-European ADS-C service offers a harmonised and simplified service interface for ground users and airspace users, shared monitoring and analysis tools, and central data repository.

- **Proven Solution**

Built on the SESAR validation and demonstrations relying on standard SWIM services for ground distribution and is air-ground radio link agnostic.

1.3 Scope of the CONOPS

1.3.1 NDTECH Prioritisation for CP1 AF6

Throughout 2021, EUROCONTROL/NM, S3JU and SDM cooperated in defining an approach for the OEP ATS B2 workstream topic and engaged with all stakeholders. Three scenarios were proposed to the NDTECH working arrangement for the ATS B2 work stream: a first 'do nothing' scenario (Scenario 1), a second corresponding to the scope of CP1 AF6 combined with a common ADS-C service (Scenario 2), a third corresponding to the use of ADS-Cv1/v2 and CPDLCv2/v3 beyond the scope of CP1 combined with a common ADS-C service and progressive deployment of ATN/IPS over new datalink technologies (Scenario 3). The description of these scenarios is attached in Annex.

NDTECH/6 of 29 March 2022 confirmed the activation of the workstream and the priority of Scenario 2 content, targeting the CP1 industrial target date, as a stepping-stone towards a future Scenario 3 (full deployment of ATS-B2). Workstream 13.10, the SWIM distribution and integration of the EPP data, is merged for efficiency reasons with workstream 12.2.

For Scenarios 2 and 3 presented to NDTECH, the need to develop a CONOPS for an ADS-C common service based on PJ38 demonstrations addressing the need for a single logon by the flight deck crew, including backwards compatibility for ground ATN B1 data link systems have been identified.

1.3.2 Controller Pilot Data Link Communications (CPDLC) v2 (or later)

CPDLC v2 (or later) is not mandated by the CP1 AF6 regulation and is therefore not developed in detail in the context of this CONOPS. However, it is important to ensure that this CONOPS document does not introduce any constraints on its future deployment, and wherever, highlight any opportunities (see sections 6.2.4 and 6.2.5).

SESAR PJ38 and PJ18 validations and current operations have demonstrated that CPDLC v2 brings benefits when it is used in combination with ADS-C. As TBO enabler, CPDLC v2 allows ATCOs to uplink complex clearances with `lat/long` coordinates and vertical constraints which can be autoloading by the pilots into the FMS. Subsequent ADS-C/EPP downlinked data can be used to check that the pilot has updated the FMS. Being able to uplink `lat/long` coordinates enables a better use of the airspace, which at the end can result in less miles flown by aircraft.

As described in section 1.3.1, CPDLC v2 may be further addressed as of 2024 (Scenario 3).

1.3.3 Global Interoperability

The CP1 AF6 regulation requires the ADS-C downlinking of the extended projected profile as part of the ATS B2 services which is detailed in EUROCAE/RTCA ED-228 [Ref 3]. The EUROCAE/RTCA ED-229 [Ref 4] interoperability standard further details the communications environment for the CM, CPDLC and ADS-C ATS B2 applications; it refers to ICAO defined ATN/OSI and foresees ICAO ATN/IPS operating over various air-ground radio links and ground-ground infrastructure.

It is to be noted that the SESAR/NEXTGEN agreed global interoperability target is the use of ATS B2 over ATN/IPS. A multilink roadmap supporting this interoperability target is not part of this CONOPS.

Furthermore, this CONOPS does not address nor impact the use of FANS ADS-C which only supports the 'Projected Profile', nor AOC applications data exchanges.

1.3.4 Costs Benefits Analysis

A full Cost Benefit Assessment (CBA) of the content of this CONOPS has not been developed, however numerous benefits are referred throughout. These benefits are qualitative and derived from validation, demonstration, operations, and feasibility. In particular, the operation of an ADS-C common service not only reduces ground investment costs but also improves spectrum efficiency as VHF band is identified as scarce resource.

It is to be noted that the expression of AF6 benefits in financial terms has been previously assessed by SDM leading to the CP1 regulation. In addition, the ATS Common Datalink Service (ACDLS) governance which aims to procure common datalink services is performing a CBA.

This CONOPS therefore builds on the identified benefits and focuses on the best technical and operational manner to support CP1 AF6 implementation.

1.3.5 End-to-End Integrity of ATS B2 Application Data Exchanges

As detailed in this CONOPS the introduction of the ADS-C Common Service impacts the end-to-end exchanges as follows:

- for the CM application a new ground facility acting as ground CM-server has been identified, the Logon Service, becoming a new end point;
- the ground ADS-C application endpoint is the ADS-C Common Service and not the user who connects to it through a SWIM service;
- the end points of CPDLC connections remain unchanged (aircraft and ACCs).

The applicability of the EUROCAE/RTCA ED-228 [Ref 3] safety and performance requirements to the Logon Service and the ADS-C Common Service will be part of the Service Assumptions deliverable (DEL2.1-2 of the topic plan [Ref 1]) and further refined in the specifications (DEL2.1-4 of the topic plan).

1.4 Operational Requirements for the CONOPS

The following operational requirements have been identified for the deployment of the ADS-C common service. These requirements have been derived from the validation, demonstration, and implementation of ADS-C/EPP data by operational stakeholders:

- 1) Single manual flight deck logon by crews covering both CPDLC and ADS-C applications
- 2) Logon as early as possible so that users can access ADS-C/EPP data (for both internal departures and incoming flights)
- 3) Minimal impact on flight deck procedures or simplification of the existing procedures
- 4) Non-disruption to existing ATN B1 fleet unable to connect to the Logon Service (i.e., non-retrofitted NSAP address list and ATC centre database)
- 5) No negative impact on the transfer of CPDLC current data authority (CDA) and next data authority (NDA) (incl. ground exchange of LOF/NAN)
- 6) No change to ground forwarding of On-Line Data Interchange (OLDI) LOF and NAN messages [Ref 9] between ACCs
- 7) Transition scenarios for non-connected ANSPs to the Logon Service (sharing of logon information)
- 8) Exchange of complete logon information between ANSPs and the Logon Service
- 9) Optional: Central and separate logon lists for CPDLC and ADS-C

1.5 Structure of this Document

The structure of this CONOPS document follows the four themes covering different aspects of the downlinking and ground distribution of ADS-C/EPP data that were identified and assessed by OEP WST 12.2 members:

- **Section 2 – Overview of the CONOPS**
Overall description and illustration of the Logon and Common Services that fall within the scope of the CONOPS prepared by the workstream.
- **Section 3 – Logon Service**
Description of the target and complete Logon Service retained for the CONOPS.
- **Section 4 – ADS-C Common Service**
Description of the ADS-C Common Service based on SESAR PJ38 [Ref 2] deliverables enhanced with additional services options.
- **Section 5 – Service Evolutions and Transitions**
Coverage and scalability of the Logon and ADS-C common services, including their co-existence with non-participating ANSPs and suitable transitions plans.
- **Section 6 – Procedures**
Impacts on procedures for airspace users, ANSPs, NM the Logon Service and the ADS-C common service.
- **Annex A**
All CM-logon options that were identified and assessed by the workstream to define the Logon Service.
- **Annex B**
NDTECH and JCSP overall scenario descriptions for the OEP ATS B2 workstream 12.2.

2 Overview of the CONOPS

This section provides an overview of the information flows addressed by this CONOPS illustrated by Figure 1.

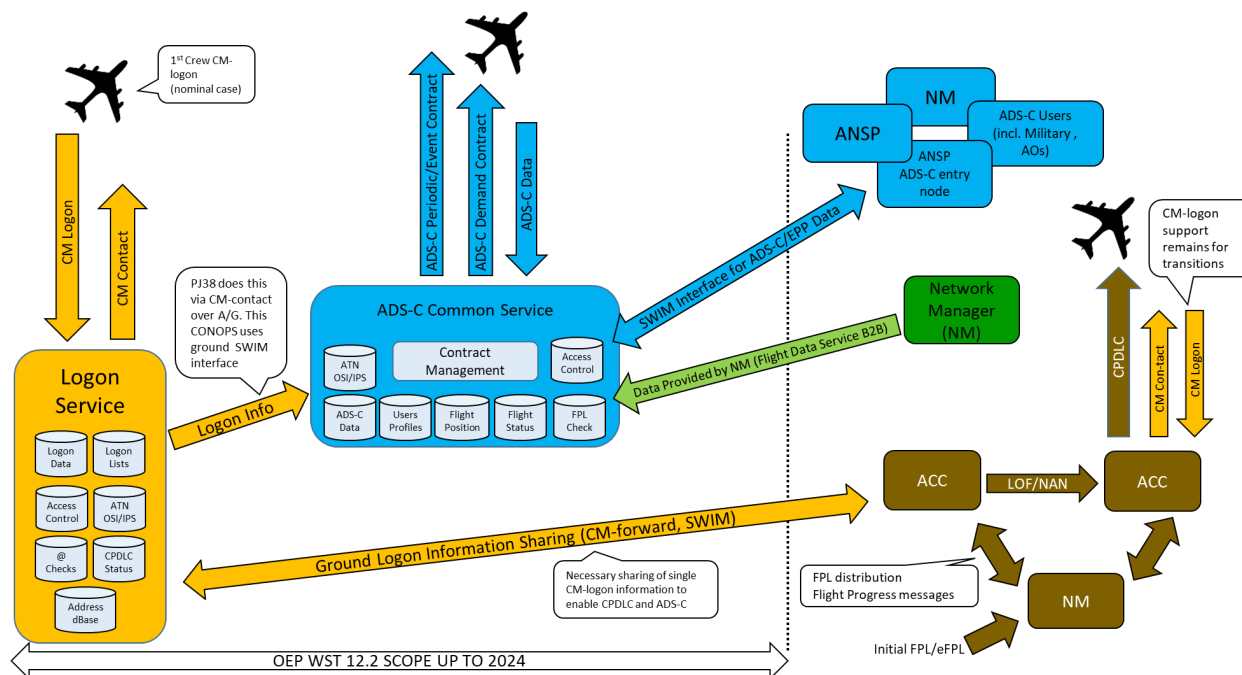


Figure 1: CONOPS Overview

In Figure 1, the Logon Service (in orange) and an ADS-C Common Service (ACS) (in blue) are functionally separated to clarify information flows. In practice, these can be merged meaning that the ADS-C common service would include the necessary logon functionalities acting as CM-server.

The Logon Service implements DLIC datalink messages (CM-logon, CM-contact and CM-forward) acting as a CM-server as specified in ICAO Doc 9880 and EUROCAE/RTCA ATS B2 standards. Primarily, it stores and shares logon information with the ACS (to establish ADS-C contracts) and ANSPs (establishing CPDLC connections). It is complemented by additional features described in section 3.4 which will be further detailed in the Specifications to be prepared by the workstream.

The ADS-C Common Service manages the ADS-C contracts with aircraft and provides a SWIM service interface so that remote ADS-C users can receive ADS-C/EPP data over ground infrastructure. The use of external flight status and positional data provided by NM (green) will improve the management of contracts as currently demonstrated by SESAR PJ38.

The following colour coding has been applied:

- Orange:** Air-ground logon information flows between the aircraft and ground facilities (CM-logon, CM-Contact) and ground-ground logon information flows between the Logon Service and the ACS/ACCs (CM-forward, SWIM).
- Blue:** ADS-C common service air-ground data flows with aircraft, and its ground-ground SWIM interface to distribute data to remote ADS-C users.
- Green:** Receipt of Network Manager data (NM B2B Flight Service) to improve the ADS-C Common Service (as used by PJ38).
- Brown:** ANSP initiated CPDLC and existing ground-ground ATS information flows (NM systems, ANSP FDP systems) that are not affected by this CONOPS.

3 Logon Service

3.1 Introduction

The datalink initiation and capability (DLIC) service is a pre-requisite for the initiation of CPDLC and ADS-C applications as detailed in ICAO/EUROCAE standards [Ref 3, 4 and 6]. The DLIC service is invoked when the flight deck crew issues a CM-logon request which indicates the supported applications and their versions. For current ATN B1 datalink services, the issuing of the CM-logon request is manually initiated by the flight-deck crew.

In the context of ATS B2 and an ADS-C common service (ACS): ANSPs need to be aware of the logon status to establish CPDLC connections and the ACS needs to be aware of the logon status to establish ADS-C contracts. Requesting the flight deck crew to manually issue two CM-logon requests is inefficient, especially when one considers that for the current ATN B1 datalink operations, not all crews perform this task.

This section details the Logon Service enabling **a single CM-logon request by the flight deck crew**. It describes the outcome of extensive work performed by the workstream in identifying, assessing, and evaluating 12 options detailed in Annex. In short, it ensures the ground distribution of CM-logon information between ANSPs, the Logon Service and the ACS.

It is to be noted that three options considered by the workstream were discarded as non-standard. In Annex, these are 'autologon' (option 10), 'forced CM-logon' (option 11) and 'ADS-C without logon' (option 12) and may be considered by ICAO/EUROCAE/RTCA/AEEC standardisation organisations. More importantly, standardisation organisations should integrate the notion of ACS and a Logon Service in their published standards and re-consider ADS-C contract prioritisation (currently the ATSU CDA). The possible future standardisation of these items does not affect the principles of this CONOPS.

It is understood that the issuing of CM-logon request to the Logon Service instead of an ACC is compatible with EUROCAE ED-120 [Ref 6] and therefore complies with the ATN B1 datalink services regulation (Commission Regulation (EC) No 29/2009 [Ref 12]).

3.2 Long-term Objective

For simplification purposes this section co-locates the Logon and ADS-C Common Services to explain the long-term objective of the CONOPS regarding logons. Figure 2 illustrates the aircraft performing a logon with the Logon Service so that ADS-C contracts can be established as soon as possible. This will meet the first three requirements of section 1.4 by allowing flight-deck crews to logon once they are within ATN coverage and the early distribution of ADS-C/EPP data by the ACS.

The ACCs are best positioned to determine when the aircraft logon information is necessary to establish CPDLC. The ACC will trigger a ground SWIM request to the Logon Service who will provide the information in return via CM-forward or SWIM.

The numbered arrows of Figure 2 are as follows:

- 1) Flight crew issues a CM-logon request to the Logon Service
- 2) ACS can establish ADS-C contracts
- 3) ADS-C/EPP downlinked data is distributed by the ACS to the ADS-C users
- 4) ANSP SWIM request to the Logon Service to receive logon data
- 5) The logon information is returned by CM-forward or SWIM
- 6) ACC establishes CPDLC connection
- 7) ACC sends OLDI LOF/NAN messages to the downstream ACC

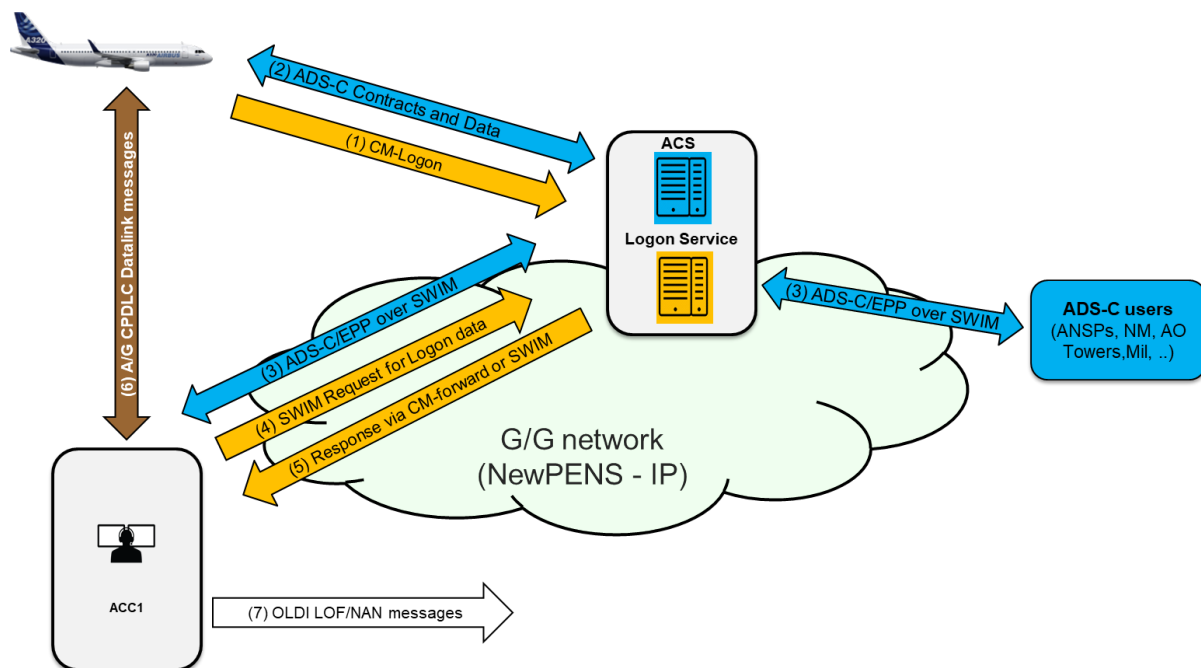


Figure 2: Objective of the Logon Service (in orange)

3.3 Logon Service

To accommodate a transition from today's environment to the long-term objective of section 3.2, numerous flows need to be implemented by the Logon Service. These flows are illustrated in Figure 3 allowing various transitions as described in section 5.6.

ATS B2 aircraft are expected to logon to the Logon Service (arrow (1)). ACCs will need to continue processing direct CM-logon requests and forward logon information to the Logon Service (arrow (9a)). The logon procedures will be documented in national AIPs, therefore the early establishment of ADS-C contracts will depend on AU capabilities and AIP instructions.

ACCs will need to forward received logon information from OLDI LOF messages (arrows (9b) and (10)) to the Logon Service. Indeed, aircraft may have already performed a logon with an upstream ACC that does not participate to the Logon Service, and this increases the resilience of information sharing between participating ACCs and the Logon Service (e.g., recovery from failures in section 5.5.2).

When ACCs are ready to process logon information for a flight, an internal trigger (arrow (4)) will generate a SWIM request to the Logon Service (arrow (5)). ACCs can indicate by which means the logon information is to be returned. To adapt to local ACC needs and transitions the returned information can be via ground-ground CM-forward (arrow (6a)) or SWIM (arrow (6b)), or even via ground-air initiated CM-contact (arrow (6c)). The ACC request will need to indicate whether the Logon Service is to respond immediately with any available information, or once information becomes available within a set time.

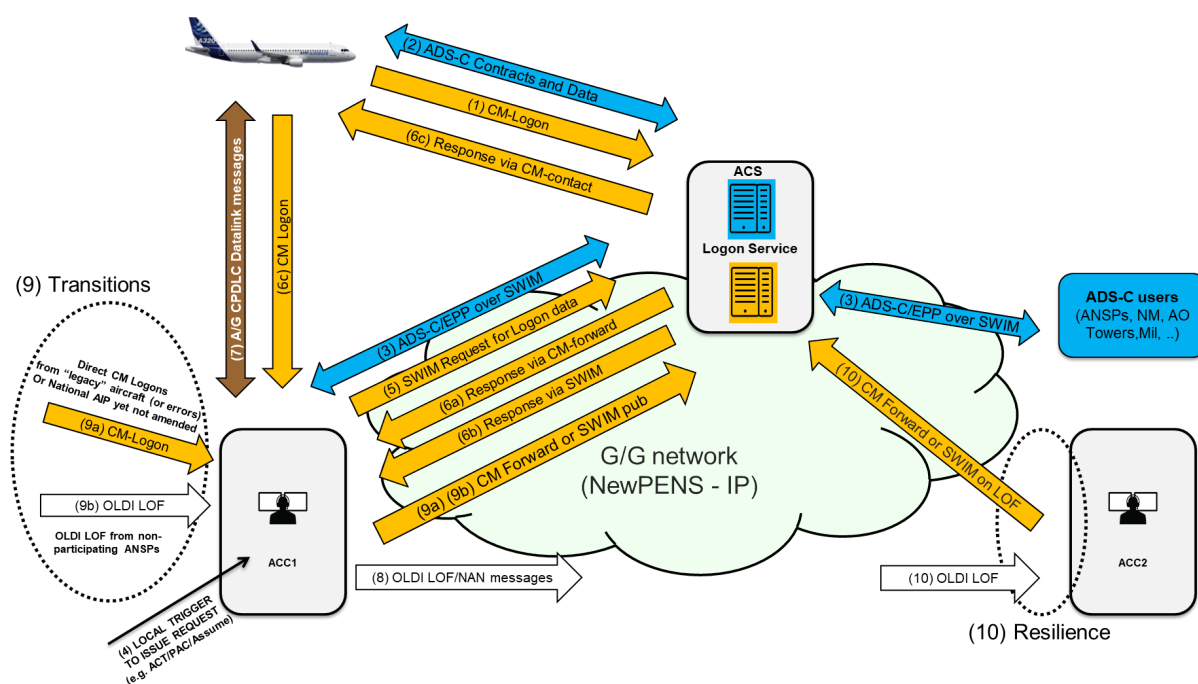


Figure 3: Complete Logon Service (in orange)

It is to be noted that to support the CPDLC application and the single-logon approach for datalink services in Europe, the ground forwarding of OLDI LOF/NAN messages must be maintained by all ANSPs.

3.4 Features of the Logon Service

A key benefit of the Logon Service is the harmonisation of logon procedures for flight crews. It aims to use the same logon address for the European region in contrast to the today's European multiple-address architecture. The administration of airborne ATC Centre databases will be simplified. Moreover, an evolution of the Logon Service should allow the flight crew to logon whilst on the ground (before take-off) or as soon as the aircraft is within ATN coverage. This would allow operators to insert the logon procedure in the routine flight crew Standard Operating Procedures / Flight Preparation, in comparison with the current random logon above FL250. This should significantly increase pilot logon rate and datalink use in Europe.

Basic and advanced features of the Logon Service are described throughout this document; however, these can be summarised as follows:

- European ground facility designator (GFD)² to receive CM-logon requests from aircraft over ATN/OSI or future ATN/IPS
- Storage of received B1 and B2 logon information
- Distribution of Logon Information
 - Provide a (SWIM) interface to receive/transmit logon information
 - Receive/transmit CM-forward messages with ACCs
 - Uplink CM-contact requests
- Managed services
 - Restricted access and authorisations for ACS, ANSPs and NM (for datalink monitoring purposes)

² Alternatively, an existing and unused address from ICAO DOC 028 could be reassigned to the Logon Service.

- Harmonised logon list(s)
- Service availability notification for ANSPs and scheduled downtimes
- Stored B1 and B2 logon information automatically overwritten and purged after end of flight or a given time (e.g., 12 hours)
- Provide a (SWIM) interface to receive/transmit CPDLC CDA/NDA status with ACCs
- Address validation of CM-logon data and automatic reporting to the Central Reporting Office (CRO)
- Prevention and detection of data exchange 'loops' with ACCs
- Compliance with applicable CM-server requirements in ICAO/EUROCAE standards
- SWAL compliance (e.g., SWAL 3 depending on the safety analysis)

A full review of relevant ED-228A [3] safety and performance requirements applicable to the Logon Service acting as ATSU CM-server will be part of the Service Assumptions deliverable (DEL2.1-2 of the topic plan [Ref 1]) and further refined in the specifications (DEL2.1-4 of the topic plan).

4 ADS-C Common Service

4.1 Introduction

ATS B2 equipped aircraft are technically capable of supporting ADS-C contracts with at least 4 ground facilities in parallel (5 for current Airbus operated implementations). A key goal of the ADS-C common service (ACS) is to avoid abusing this capability, downlinking several times the same (or similar) information, regarding the scarce resources of the VDLm2 air-ground radio link.

The ADS-C common service assumes responsibility for collecting ADS-C data from airborne users and distributes this data to entitled ground users. The overall objective of the ACS is to establish and maintain an ADS-C connection with an aircraft as early and as long as possible, covering all phases of its flight. The ACS will collect data whether its departure (ADEP) and/or arrival (ADES) lies outside or inside the area covered by the service.

The ADS-C common service covers:

- the **collection** of ADS-C data from the aircraft via a single ATS B2 ADS-C application dialogue to reduce ATN air-ground network load and use of radio spectrum.
- the **distribution** of data to clients via SWIM ground distribution services (AMQPS and HTTPS technologies)

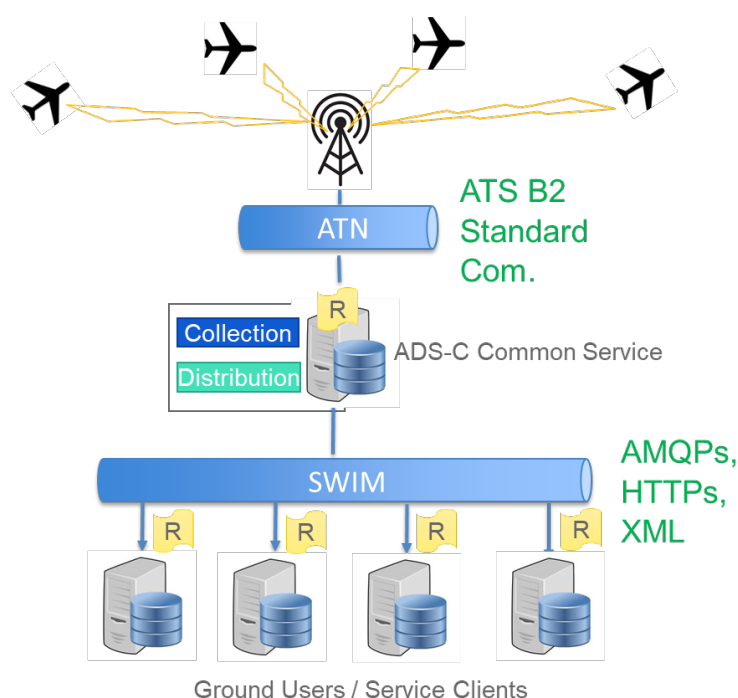


Figure 4: ACS Overview

To limit the number of simultaneous contracts to obtain the same ADS-C data, the ADS-C common service shares contract definitions among its users.

In general, this means that the ACS will establish one periodic and one event contract per aircraft flight for all its users. In addition, the service also supports ADS-C demand contract establishments to meet specific user data requests.

4.2 Service Assumptions

The assumptions for the ACS are that:

- aircraft logon information (ATN addresses, applications, and versions) is made available to the ACS, via a SWIM interface to the Logon Service, to enable the establishment of ADS-C contracts
- standard contracts cover most user needs, most of the time
- standard contracts will be established to request all data elements and events that are not situation dependent
- the service aims to establish contracts with all ATS B2 equipped flights operating in SES airspace
- an increasing number of aircraft will be ATS B2 equipped with ADS-C v1 capabilities in accordance with the CP1 regulation
- ground operational users accessing the service, such as ANSPs, may or will need to receive ADS-C data beyond their operational area of interest

4.3 Users of the Service

In accordance with the CP1 regulation, ANSPs and NM must make use of the downlinked ADS-C/EPP data. However, additional users would also benefit from the data further justifying the rationale of an ADS-C Common Service (e.g., airline operators for fleet management, airport operators for traffic planning, meteorological services to obtain airborne weather data, military organisations performing ATM functions). For each user, there are several use cases for the data (FDPS conformity monitoring, FDPS MTCDs, Arrival Management, ATCO CWP information display, etc.).

User access to the service will depend on a service delivery agreement covering the authentication requirements, authorised data access, and terms and conditions on the use of the data.

4.4 Service Functional and Operational Description

4.4.1 Service Characteristics

- Periodic and event contract definitions and parameters will be agreed in advance by appropriate governance representing the users, with the goal of maximising benefit to the user community while respecting performance limits of the network.
- Users will be known before requesting the service (user registration prior authentication)
- Users will have different and limited access rights in accordance with their service delivery agreement (e.g., some users may only be entitled to see subsets of flights, airlines for their own fleet).
- The access to demand contracts will be limited to users with certain privileges.
- Users and privileges will need to be managed.
- Between the subscription and publishing of data a process will need to ensure that the user is entitled to this data
- The frequency of demand contract requests of one ground-user per aircraft will be limited (e.g., to 1 request per minute, configurable)
- All downlinked ADS-C/EPP data distributed to users is encoded using ASN.1 XML Encoding Rule (XER)

- The ACS will attempt to recover from any aborted ADS-C dialogues considering any known capabilities of the concerned aircraft or reported avionic issues (such as those experienced during SESAR PJ38 demonstrator)
- ACS stores ADS-C data (event and periodic) so that users connecting for the first time can request flight status and the last report

4.4.2 Air-Ground Data Collection

4.4.2.1 Receipt of CM-logon Information

For the ACS to establish contracts with the aircraft, it needs to receive logon information from the Logon Service. This is enabled via a SWIM interface between the ACS and the Logon Service.

4.4.2.2 ADS-C Contract Management

The ADS-C service will need one registered ground facility designator (GFD) to support the establishment of periodic, event and demand contracts. A key principle for the management of periodic and event contracts will be the independence and decoupling of ADS-C Contract Management from the user access to the service.

4.4.2.2.1 Periodic and Event Contracts

As soon as the ACS receives logon information for a given flight, it will attempt to establish one periodic and one event contracts with the aircraft in accordance with pre-defined 'standard' contract elements.

In the event of notified provider abort, the ACS will attempt to re-establish prior contracts (see section 6.4.2).

4.4.2.2.2 Demand Contracts

Demand contracts allow access to data elements that are not covered by the standard periodic and events contracts or when the ACS data is not sufficiently recent.

Users of the service can request the ACS to establish a demand contract for a given flight as long as the request relates to data elements defined by the ATS B2 standard. This request is enabled via the SWIM interface to the ACS and will lead to a new ADS-C contract with the aircraft.

The demand contract will be closed as soon as the requested data elements are received by the ACS. The received data is then distributed to the user over the SWIM interface. Any received data will also be distributed to other users, provided they have a subscription for ADS-C data from this flight.

When a user requests a demand contract while another demand contract for this aircraft is being processed, the service will reject the request. Indeed, only one demand contract request can be invoked per aircraft at a given time. The user will need to request such rejected demand contracts later if necessary.

4.4.2.3 Use of Aircraft Position Information

The ADS-C service will be improved by being connected to a source of aircraft position information. This will be provided by the NM B2B Flight Data Service as currently used by PJ38 (subscription to the B2B "Flight Services" which publishes the `lastknownPosition`). If external position data is missing (source temporarily unavailable, or no position data for a

specific flight) ACS could still apply default contracts or work with the less frequent position updates included in ADS-C reports.

When available, the aircraft position information will be used to enhance the management of ADS-C contracts:

- Identification of aircraft entry and exit to the area covered by the service
- Timely triggering of ADS-C periodic and event contracts
- Handling of demand contracts (checking that the aircraft is within the area covered by the service)
- Modification of the ADS-C contract based on the aircraft's 3D position (location and phase of flight)

4.4.2.4 Use of Variable Periodic and Event Contracts

The ADS-C service could provide improved services for users by establishing tailored periodic and event contracts based on flight characteristics, instead of using standard definitions.

Some of these flight characteristics depend on the access to other data sources such as aircraft position information (see section 4.4.2.3) and include:

- the indicated ADEP/ADES in the received logon information,
- the specific airspace in which the flight is operating, which can correspond to a flight segment,
- the phase of flight,
- waypoints or calculated points (e.g., ToC, ToD) in the received EPP

This means that flights will be monitored by the ACS, and depending on the flight characteristics, existing periodic and event contracts can be replaced by others.

As this provides additional functionality to users which can be considered as an operational benefit, it will be brought forward as a recommended requirement within the service specifications.

4.4.2.5 Use of Flight Status Data to Manage Contracts

The ADS-C service will be improved by being connected to the NM B2B Flight Data Service as a source of flight status data (currently used by PJ38 platforms).

When available, the ACS will use flight status data to enhance the ADS-C contract re-establishment. Indeed, in cases of provider aborts and as long as the flight is active in the NM system, the ACS should try to re-establish prior ADS-C contracts with the aircraft.

The general principle here is to use NM B2B data as a complementary source of information to manage ADS-C contracts.

4.4.3 Ground Data Distribution

The service provides a ground data distribution interface compliant with the SWIM yellow profile service specifications.

4.4.3.1 Data Distribution / Service Interface to Users

Users connect to the ACS in accordance with the SWIM interface and are subject to authentication and authorisation mechanisms.

Users subscribe to CM and ADS-C datalink messages published and made available by the ACS. The subscription may be customised at least by:

- a) message type,

- b) Flight ID (All or specific flights),
- c) ADS-C Reports from aircraft with a verified flight plan (option – see section 4.4.3.3)

Following user subscription, the ACS will need to perform an additional check of the user's privileges to access the requested data (access rights vs subscription).

Table 1 indicates how the needs of key users accessing the ACS (ANSPs, NM, and AUs) can be translated into subscriptions and their associated parameters. These parameters will be defined in the SWIM ground distribution service specification (DEL2.2-1 of the topic plan [Ref 1]).

Subscription examples	ANSP	NM	Airspace Users
Flights of Interest	All flights planned to cross ANSP Area of Interest	All flights	Fleet or aircraft, based on ICAO 24-bit address
Selection criteria for flights	Per flight, based on Callsign, ADEP, ADES, EOBT and EOBD, icao24bit address FPL verified flights-only	All flights	Permanent subscription for fleet e.g., based on list of icao24bits or Airline Code in Callsign Combination of "ICAO Airline Identifier" (Three characters) and icao24bit address. (logical AND) FPL correlated flights-only
Data of Interest	All Messages (including CM) Individual ADS-C Message types to be analysed (make configurable) Only logon and ADS-C Reports? Or also other status information, Contract Requests, Aborts....	All Messages (including CM)	ADS-C Reports
Request to establish demand contracts	Yes	Yes / Potentially	No (t.b.c)

Subscription examples	ANSP	NM	Airspace Users

Table 1: ADS-C User Subscription Examples

4.4.3.2 Requesting Flight Status, Aircraft Status and Prior Reports

The ADS-C common service will provide its users with information describing the status of the ADS-C connection of individual flights or aircraft. The status information is made available either through a SWIM compliant request-reply interface or as metadata attached to the subscriptions established by the user.

The status information available through both request-reply and subscription metadata encompasses:

- The time of the last ATS B2 ADS-C message successfully exchanged with the aircraft,
- The logon status of the aircraft (logon received or not) including if applicable the result of the flight plan verification (see section 4.4.3.3),
- The status of the periodic, event (established or not) and any active demand contract.

The information only available through request-reply includes (while an ADS-C dialogue is established):

- The periodic, event and any demand contract definitions currently applied to the aircraft,
- If any, the last ADS-C report received from the aircraft,
- If any, the time and type of the last user or provider abort received from the aircraft
- If any, the status of any pending connection retries being performed by the ACS contract management logic.

4.4.3.3 Verification of Flight Plan Data for Received Logons

The ADS-C service could be improved by being connected to the NM B2B Flight Data Service as a source of flight plan (FPL) data (currently used by PJ38 platforms).

When available, the ACS will verify received logon information against NM FPL data. The possible outcomes of the verification are 'FPL verified', 'FPL inconsistent' or 'FPL inexistent' and will be communicated to the users. It remains in the responsibility of the user to decide whether such data will be used.

Irrespective of the verification process outcome, the ACS service will establish ADS-C contracts with the aircraft.

The service will provide means to subscribe to data based on the outcome of the verification process. It is to be noted that the ACS verification does not relieve the ANSPs from performing their local FPL correlation as this is a safety requirement of the ATS B2 standards.

As this verification is informational, it will be brought forward as an optional requirement within the service specifications.

5 Service Evolutions

5.1 Introduction

The CP1 regulation mandates ANSPs³ to support the reception and processing of trajectory information (ADS-C/EPP data) from aircraft operating as general air traffic above flight level 285 (FL285). As strongly recommended by the SESAR Deployment Programme, the ADS-C Common Service has been identified to enable reliable, fast and efficient initial trajectory information sharing for ANSPs. In this manner, ADS-C/EPP data can be made available and used by all ANSPs, NM and other ADS-C users that are part of the ACS, irrespective of which ANSP is effectively delivering control services to the flight.

Initially, the ACS coverage area will be defined in accordance with the needs (and area of interest) of the participating ANSPs, however, it offers a clear path to extend ADS-C coverage to all SES airspace and beyond, and even below FL285.

ANSPs that are not part of the ACS may be adjacent to the ACS coverage area or even within the coverage area creating discontinuity:

- When an ADS-C service is not available in a given ANSP (because it is not part of CP1 or the implementation has been delayed), the ADS-C/EPP data will not be used by that ANSP, but the ACS contract, if already active, will be kept. The ADS-C/EPP data can continue to be distributed to ANSPs, NM and other ADS-C users that are part of ACS.
- When there is an ADS-C service in a given ANSP which is not part of the ACS, it is highly likely that the ANSP and ACS will have separate simultaneous ADS-C contracts with the aircraft. This fragmented approach is not desired due to the increased use of air-ground and airborne resources (radio spectrum, ATN VDLm2 network load, number of simultaneous contracts).

In any event, there will be a period (transition period) where not all ANSPs will be part of the Logon Service and/or the ACS. During that period, a mixed scenario and transition plans need to be considered as addressed by this section.

5.2 Coverage Area

5.2.1 Initial CM-logon Request Outside of the ACS Coverage Area

When an aircraft performs a CM-logon request to an ACC which is not part of the Logon Service, ADS-C/EPP data will not be available from the ACS as it is not in possession of the addressing information to initiate the ADS-C contract. It is when the OLDI LOF message is received by the first ACC part of the ACS, or when the aircraft performs a new logon either to the Logon service or to an ACC part of ACS, that the addressing information can be made available to the ACS to establish an ADS-C contract.

To accelerate the establishment of ADS-C contracts by the ACS, the use of position and flight status information combined with a database of previously received addressing information could be used. This mode of functioning is further described in section 6.4.3 'Procedure to Increase Logon Connection Rates'. Alternatively, as part of a bilateral agreement, the ACC adjacent to the ACS coverage area could forward any logon information directly to the Logon

³ All ATS units providing air traffic services within the airspace for which the EU Member States are responsible in the ICAO EUR region and ANSPs of non-EU Member States that apply SES regulations. By extension the Logon Service and ACS will be open to any ANSP that is not subject to the regulation.

Service before LOF transmission to the downstream ACC. In both cases, the aircraft must be within the ATN coverage for the ACS to establish ADS-C contracts.

5.2.2 Discontinuity within the ACS Coverage Area

Once the ADS-C contracts have been established by the ACS, these can be maintained even when the aircraft flies into airspace of an ANSP that is not connected to the ACS (or not providing ADS-C services). Indeed, based on flight plan data provided by NM B2B Flight Data Service or even EPP information, the ACS will determine that the aircraft will re-enter airspace of an ANSP that is connected to the ACS and therefore maintain the contracts.

5.2.3 Aircraft Leaving the ACS Coverage Area Permanently

Based on flight plan data provided by NM B2B Flight Data Service or even the trajectory data provided by EPP information, the ACS can determine an aircraft is leaving the ACS coverage area permanently. The ACS should terminate the contracts.

5.2.4 ANSPs Part of ACS but not the Logon Service

Some ANSPs may not be part of the Logon Service and just part of the ACS. Indeed, the implementation of a ground SWIM interface to the ACS may be beneficial for ANSPs that are not subject to the CP1 regulation or in the process of migration (e.g., for test and evaluation purposes, for ATCO training).

However, such arrangements are undesirable for the long-term as it will impact the ACS access to ADS-C EPP data. Indeed, ADS-C contracts cannot be established by the ACS for flights departing within the area of those ANSPs until the flight crew decides to logon to the ACC, a LOF message is sent to the downstream ACC and then forwarded to the Logon Service (refer to Figure 3).

5.3 Backwards Compatibility

5.3.1.1 Mixed Fleet and Procedures

The CP1 AF6 regulation, requires aircraft with an individual certificate of airworthiness first issued on or after 31 December 2027 operating above FL285 to be ADS-C EPP equipped. This means that from 2028 onwards, aircraft operating above FL285 will be aircraft with a CoA before 31/12/2027 (ATN B1 equipped, ATN B1 exempted and ATS B2 equipped) and new ATS B2 aircraft with a CoA after 30/12/2027 (at least CPDLCv1 and ADS-C EPP equipped).

Until the putting into service of the Logon and ADS-C Common Services, all aircraft will continue to log on to ACCs whether they are ADS-C EPP capable or not. Once these are put into service, updated AIP procedures for datalink services will need to be published. Aircraft will be instructed to logon to the Logon Service instead of the ACC. However, as described in section 3.3, ACCs will need to retain the capability to accept direct logons from aircraft. For example, this would be to support aircraft without the configured Logon Service GFD, mis-directed logons or non-updated flight deck procedures. As a result, the ground CM entity (ACCs or Logon Service) to which the aircraft will perform the initial CM-logon will vary over an extensive period depending on capabilities and procedures.

5.3.1.2 ACC Forwarding Requirement for ATS B2 Aircraft Logon Information

ACCs will require the capability to forward received "B1+B2" logon information to the Logon Service so that the ACS can establish ADS-C contracts. This logon information can be received from a direct CM-logon request or an incoming OLDI LOF message. Depending on the

capabilities of the ACC, the ground forwarding of this logon information will be via CM-forward or SWIM.

5.3.1.3 ACC Forwarding Requirement for ATN B1 Aircraft Logon Information

As ATN B1 aircraft do not support ADS-C, the need to forward any received “B1-only” logon information to the Logon Service is not required by the ACS. However, the forwarding of this information independently of any applied logon list could enhance the role of the Logon Service as a central repository of addressing information accessible to all ANSPs.

Accessibility to B1-only aircraft addressing information will be improved by forwarding any received ATN B1 CM-logon information to the Logon Service. Depending on the capabilities of the ACC, the ground forwarding of this logon information will via CM-forward or SWIM.

Applying the same forwarding mechanisms for “B1-only” addressing information as for “B1+B2” will also simplify ANSP logon information forwarding logic to the Logon Service.

5.4 Scalability

As described above, the expectation is to cover all SES airspace and beyond. This will be gradual and is achieved by decoupling ATS B2 CPDLC and ADS-C application data flows with the deployment of the ADS-C common service. Appropriate service governance should be established to decide on a minimum number of participating ANSPs and the initial service area. This can be followed by horizontal and vertical expansion to avoid or minimise service discontinuity.

As highlighted in section 1.2.3, the number of users connecting to the ACS to receive ADS-C/EPP data does not impact ANSP core processes nor add to air-ground datalink exchanges. Their number does not impact the scalability of the service.

ANSPs and NM can decide when to participate to the Logon Service and/or ACS depending on local deployment strategies and readiness. It is recommended that ANSPs and NM subject to the CP1 AF6 regulation connect to one or both services to meet the regulatory implementation dates.

Therefore, the factors that may constrain scalability could be the adopted service business model, the deployment of the Logon Service and ACS, delayed sharing of logon information, heavy procedures for AUs and ANSPs, and the ATN network coverage and capacity.

5.5 Resilience

5.5.1 Overview

When the flight deck crew issues a CM-logon request to the Logon Service, there is a clear dependency on its availability so that ANSPs can perform CPDLC, and that the ACS can establish ADS-C contracts. On the other hand, when the flight deck crew issues a CM-logon request to the ACC and logon information is forwarded to the Logon Service, there is a clear dependency on the ACC to handle such logon requests and forward the information to the Logon Service so that the ACS can establish ADS-C contracts.

These services must be failure-resilient; therefore, the service specifications will need to specify maximum resilience of the Logon Service and the ACS (e.g., maximum acceptable failure time per year, maximum number of occurrences) and the reliability of their means of interconnection over NewPENS. The below sections identify the impacts of their failure and mitigating options.

5.5.2 Logon Service Failure

For aircraft that have already performed the logon process, a failure of the Logon Service will not affect the CPDLC application which depends on the OLDI LOF/NAN ground exchanges nor affect ADS-C/EPP data downlinking of existing contracts that will be maintained. It is only in cases where flight crews perform a logon after the Logon Service failure that this becomes apparent.

- If flight crews logon (or re-logon) to the Logon Service, service failures will be detected as CM-logon is a confirmed service. A mitigating option would be that the crew performs a logon to the controlling ACC (this procedure can be detailed in the AIP or instructed by the ATCO).
- If flight crews manually logon or re-logon to the ACC, the Logon Service failure will not be detected. The ACC attempt to forward logon information to the Logon Service will fail.

By ensuring each ACC forwards any received logon information from a direct CM-logon request or received OLDI LOF message as detailed in section 3.3, multiple attempts will be made by ACCs to forward logon information to the Logon Service as the flight progresses. This will facilitate ACS ADS-C contract establishment for new flights once the Logon Service recovers.

5.5.3 ACS Failure

All ADS-C users depend on the availability of the ACS to receive ADS-C/EPP data. Failure of the ACS will be notified to (or detected by) all ADS-C users via the SWIM service interface.

No generic mitigation can be foreseen as most ANSPs connected to the ACS will not deploy the ADS-C contract management application at their centres.

It is to be noted, however, that ANSPs that have logon information, are controlling the aircraft and have implemented the ADS-C contract management, may decide to establish an ADS-C contract. A harmonised approach should be recommended in the guidance material to avoid excessive and simultaneous ADS-C contract establishment requests by such capable ANSPs in case of ACS failure.

5.5.4 Mis-directed CM Logon Requests

In a context of transition, a mixed fleet and procedures described above, it is possible that the flight crew issues a CM-logon request to the wrong ground facility such as issuing the CM-logon request to an ACC instead of the Logon Service. In this case the ACC can reject the request, make use of CM-contact, or accept the request and forward the received logon information to the Logon Service. The latter is the preferred method as described in section 3.3.

5.6 Approaches for the Transition Period

5.6.1 Overview

The long-term objective of the Logon Service is that all aircraft will logon to the Logon Service so that the ACS can establish ADS-C contracts as early as possible for all ADS-C users, and that complete logon information is made available to all ANSPs operating CPDLC. The multiple features of the Logon Service and ACS will support airborne and ground actors in achieving this target over a transition period.

This implies necessary actions for ANSPs to interface with the Logon Service, AUs to establish the initial CM-logon with the Logon Service and ANSPs to participate in the enlargement of the ADS-C service area. This section will further consider the key transitions from the perspective of the ACS, ANSPs, AUs, and the network.

5.6.2 ACS

From the perspective of the ACS, service area and homogeneity are essential as outlined in sections 5.1 and 5.2. The participation of ANSPs to the Logon Service and/or the ACS has a direct impact on when ADS-C/EPP downlinked data is available to be distributed to users. Furthermore, the ATN coverage/availability is also a constraining factor. This is illustrated by the three use-cases in Figure 5, which highlight the impacts on ADS-C contracts established by the ACS as a flight progresses.

ADS-C/EPP data can only be distributed to ground users (green segments in Figure 5) once the ACS can establish and maintain a contract within the coverage area based on logon information made available by the Logon Service (received directly via CM-logon or indirectly via ACC ground forwarding).

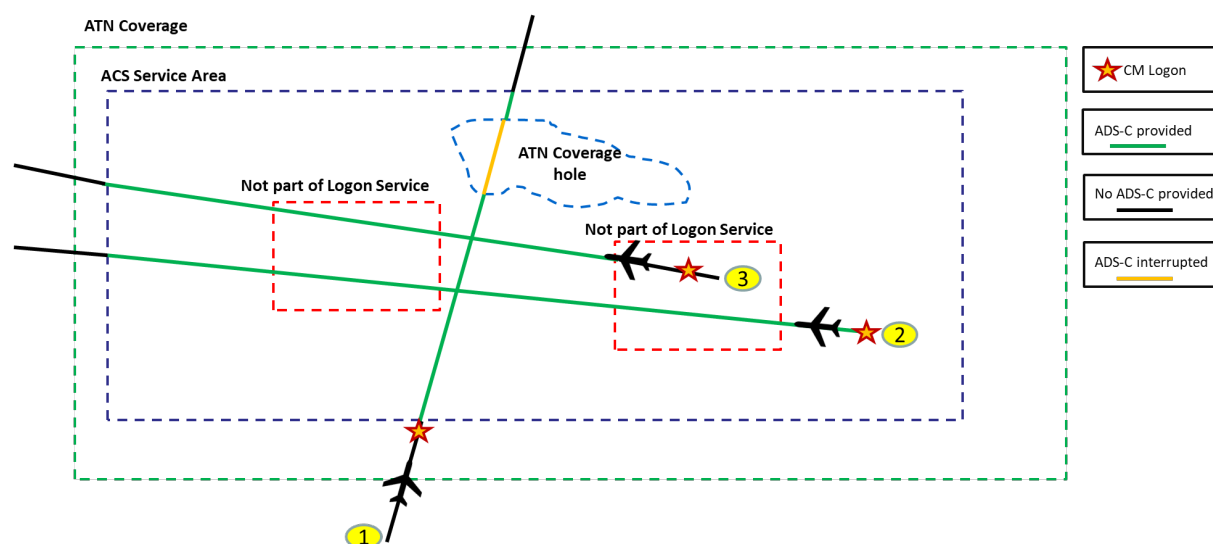


Figure 5: Service Evolution

The deployment of a homogeneous service needs to be planned with participating ANSPs, otherwise there are clear risks of simultaneous ADS-C contracts with a given aircraft and absence of data during an aircraft's flight. Furthermore, the ACS service area needs to be sufficiently large to encourage its use and to justify the transitions to be made by AUs and ANSPs.

5.6.3 ANSP ACCs

From the perspective of ANSPs, a number of steps are necessary to transition towards the use of the Logon and ADS-C Common Services.

Indeed, existing ACC CM-logon connection management needs to be reviewed and ADS-C/EPP data needs to be integrated. The Logon Service provides multiple methods to share logon information (CM-forward, CM-contact, SWIM) with ACCs so that ANSPs can select the most suitable method. ADS-C/EPP data integration will be associated with significant changes to communication systems, ATM systems and controller working positions.

As an example, the following transitional steps can be envisaged for an ANSP ACC:

- Setup and early transitions
 - Step 1: the Logon Service is put into service and ready to deliver services to ANSPs and the ACS
 - Step 2: the ACC maintains flight crew logon to its local datalink system (as for CPDLC today)
 - Step 3: the ACC can start forwarding any received logon information (CM-logon requests or OLDI LOF) to the Logon Service by making use of CM-forward or the SWIM interface
 - Step 4: the ACS is put into service and capable of establishing contracts with aircraft based on logon information forwarded to the ACS (a sufficient number of participating ACCs)
 - Step 5: the ACC becomes a user of the ACS to receive downlinked ADS-C/EPP data over the available SWIM interface
- Achieving the target
 - Step 6: the ACC implements the SWIM request interface including the preferred response method (e.g., CM-forward, SWIM, CM-contact)
 - Step 7: ANSPs and national authorities change the AIPs to indicate that airspace users need to issue CM-logon requests to the Logon Service
 - Step 8: the ACC maintains the capability to process directly received CM-logon to accommodate aircraft that are not able to logon to the Logon Service (resilience, backwards compatibility, non-updated flight deck procedures) or even mis-directed logons

5.6.4 AUs

From the perspective of AUs, instructing crews to logon to the Logon Service could depend on the scale and the coherence of the coverage area.

In practice, national AIPs will indicate to whom flight crews need to address the CM-logon request, but these may include transitional measures and may even temporarily instruct AUs to logon locally and to the Logon Service. These measures may be necessary to accommodate fleet capabilities in terms of backwards compatibility or the harmonisation of AU flight deck procedures.

Airspace users will need to take actions to update the airborne ATC centre database with the “new” Logon Service address⁴. As this can take time, a period needs to be foreseen for existing ATN B1 and ATS B2 fleet to be updated. Therefore the “logon to the first ACC” procedure will need to remain in parallel at least for a transitional period which needs to be reflected in the national AIPs. The notification procedure of the Logon Service address shall be initiated at the earliest possible time or as per AIP procedures.

The introduction of the Logon Service and ACS will also need to address existing procedures regarding re-logons, CPDLC and ADS-C re-establishment. These will be addressed by the guidelines or targeted publications to be developed by OEP Workstream 12.2 (DEL2.3-1 and DEL2.3-2 of the topic plan [Ref 1]).

Crew procedures and trainings need to be adapted accordingly.

⁴ Alternatively, an existing and unused address from ICAO DOC 028 could be reassigned to the Logon Service.

5.6.5 The Network

From the perspective of the network, the Network Manager will be significant user of the ADS-C common service by receiving the data for all aircraft. However, many other users are foreseen to receive ADS-C/EPP data (e.g., airspace users, military, weather services).

Although, no technical or operational barriers prevent serving a large number of ground users, the planning and validation of data access, system resources and network throughput need to be managed.

6 Procedures

6.1 Airspace Users

This section highlights the flight deck procedures relevant to the introduction of ADS-C and the ADS-C common service.

Airspace users shall ensure that the ADS-C function is in “armed” condition prior to the first manual logon and during the entire flight, until end of flight. Armed means that the system accepts ADS-C contract requests. ADS-C contracts are initiated from the ground and do not require further crew action. Where implemented, ATSUs holding active contracts are visible to the flight crew.

ADS-C contracts are per definition managed by the ground. Current airborne implementations allow the pilot to manually terminate the contract. In the case of a manually flight crew-initiated ADS-C contract termination, procedures need to be defined on how to re-establish the contracts. Preferably, this would be through a new CM-logon request to the Logon Service, restarting the ADS-C service, even though CPDLC exchanges may exist in parallel.

In view of the Logon Service described in section 3, two key scenarios for AUs can be envisaged:

- AU Scenario 1: National AIPs requiring a flight deck manual logon to the first ACC. Existing logon procedures apply.
- AU Scenario 2: National AIPs requiring a flight deck manual logon to the Logon Service instead of the first ACC. In this scenario the likely objective for airspace users will be that both B1 and B2 aircraft are handled equally, meaning that logon services for all aircraft should be provided by the Logon Service.

The long-term objective of the Logon Service meets the AU preferences as it envisages a single logon address simplifying the management of the airborne ATC centre databases.

The work on this CONOPS deliverable has concluded that CPDLC procedures for the flight deck remain unchanged.

6.2 ANSPs

6.2.1 Publishing Changed Procedures

The Logon Service will offer a common resource to better manage the aircraft logons, logon monitoring, logon information dissemination and a basis to support future TBO services. A key principle is that ANSPs are to forward any received logon information to the Logon Service and request data when required to perform CPDLC.

Section 5.6 provided a description of transitional steps that ANSPs should consider for interworking with the Logon Service and ACS. Such steps must be part of the overall ANSP deployment project to receive and process ADS-C/EPP data in accordance with CP1 AF6.

The deployment of the Logon Service and ACS will depend on new collective governance arrangements to setup and deliver the services. It is likely, that this will entail collective procurement actions and proper use of NOTAM and AIP publications to ensure AUs are aware of any changes.

6.2.2 Procedures relating to Systems

The integration of new SWIM interfaces to receive and process ADS-C/EPP data, the upgrading of controller working positions and ATCO training likely outweigh the efforts to deploy the new interfaces to the Logon Service.

ANSPs systems need to be modified to be able to cope with the new CM-logon logic, ADS-C contract logic, retrieve the ADS-C/EPP data from the ACS and adapt all controller positions to display trajectory information and warn of discrepancies.

ANSPs systems will retrieve ADS-C/EPP data from the ACS via SWIM standardised interfaces, therefore it would be efficient for ANSPs to expand the use of SWIM services for other use-cases described in this CONOPS. As described in section 3, ANSPs will need to act on incoming LOF messages by forwarding received logon information to the Logon Service. They will also need to maintain the capability to process direct receipt of CM-logon requests and forward this information to the Logon Service.

ANSPs should not inhibit the forwarding of received logon information to the Logon Service if the incoming LOF message is to be unacknowledged. It is possible that duplicated logon information will be exchanged between the Logon Service and ANSPs systems which needs to be handled.

As a result, multiple ANSP systems will be impacted: controller positions, binding of OLDI messages to new information exchanges, datalink systems managing CM logic, existing SWIM-based systems and communications systems. A full review of ANSP-relevant ED-228A [3] safety and operational requirements needs to be performed⁵ within the context the Logon Service and ADS-C Common Services. For example, a number of safety requirements with the SR-GD-ADSC prefix are relevant to the ANSP controller/responsible person and may need to be shared with or delegated to the ACS acting as ATSU (e.g., SR-GD-ADSC-29A). This review will be part of the Service Assumptions deliverable (DEL2.1-2 of the topic plan [Ref 1]) and further refined in the specifications (DEL2.1-4 of the topic plan).

Furthermore, clear guidance to ANSPs is to be developed indicating how systems make use of ADS-C/EPP trajectory reports in terms of their age, for the purpose of conformance monitoring and the display of information to the controller.

6.2.3 Procedures to Handle ADS-C/EPP Data

All procedures related to the CM Logon rely on automation, and as such are transparent for the air traffic controllers (ATCOs).

The ATCOs shall be made aware when the ADS-C/EPP data is available and provided for a flight under their control. Typically, this will be a specific controller working position (CWP) notification, display or colouring scheme. The ATCO shall also be made aware when existing contracts with aircraft under their control are aborted or ADS-C/EPP data is no longer available. In this regard, a number of safety requirements from ED-228A [3] with the SR-GD-ADSC prefix are relevant.

⁵ Revision B of ED-228 may reflect the existence of an ACS regarding the implementation of these requirements

The ATCO should have the possibility to request additional data and update contract parameters within the bounds of the agreed ACS service definition. These can be in the form of variable contracts (section 4.4.2.4) or demand contracts (section 4.4.2.2.2).

The following operational functionalities are mandated for those ANSPs that are subject to the CP1 regulation and strongly recommended for other ANSPs receiving ADS-C/EPP data:

- ATCOs must be able to display the route of the downlinked trajectory on the controller working position.
- Warnings shall be provided to the ATCOs in case of trajectory discrepancies between the downlinked aircraft trajectory and the ground system trajectory elaborated using the filed flight plan route.

The operational use of the ADS-C/EPP data will be further described in the future guidelines for each of the potential use-cases, to support and harmonise local implementations (DEL2.3-1 of the topic plan [Ref 1]).

6.2.4 Procedures Relating to CPDLC

The seamless operation of controller pilot datalink communications (CPDLC) depends on the:

- ground forwarding of OLDI LOF and NAN messages between the transferring ATSU (T-ATSU) and the receiving ATSU (R-ATSU), and
- uplinking datalink messages (UM183 CURRENT DATA AUTHORITY xxx and UM160 NEXT DATA AUTHORITY xxx) indicating the CPDLC current (CDA) and next (NDA) data authorities.

Most ATSUs have experienced rejected CPDLC start requests, notified by downlinked datalink message DM107 NOT AUTHORIZED NEXT DATA AUTHORITY. Several causes can explain these events such as an uplinked NDA to the aircraft sent too late or the R-ATSU attempting to establish CPDLC too early. As access to logon information is a prerequisite for the R-ATSU to be able to perform a CPDLC start request, the impact of the Logon Service needs to be considered.

The Logon Service allows the R-ATSU to receive aircraft logon information independently of the OLDI LOF message exchanges.

As a result, clear guidance to ANSPs shall indicate that the R-ATSUs can request or may receive logon information at any time, however:

- if a LOF message is not received in due time, the R-ATSU⁶ should then request logon information from the Logon Service
- the R-ATSU has to wait on NAN reception before performing a CPDLC start request (NAN reception implies the aircraft knows the R-ATSU is the CPDLC NDA)

Timing remains important as the CPDLC start request may still fail if R-ATSU is not identified as CPDLC NDA.

6.2.5 ACC Forwarding Requirement for CPDLC CDA/NDA Status

Once an ACC becomes the CPDLC CDA it can notify via the SWIM interface that it is CPDLC CDA and notify which ACC is the CPDLC NDA after it has an uplinked NDA message (UM160 NEXT DATA AUTHORITY xxx) acknowledged by a LACK. In case the CPDLC CDA notifies the Logon Service about the NDA, the Logon Service has to be notified about any changes to the

⁶ This timing can be specific to a given boundary between ATSUs and part of the bilateral agreements which are usually linked to OLDI notification (ABI) and/or coordination (ACT) message reception. A limited number of CPDLC start re-tries should be attempted.

NDA (e.g., in case of re-routings). In addition, the CPDLC CDA could also notify the Logon Service about the termination of its CPDLC connection.

Two key benefits associated with this information sharing have been identified as follows:

- a receiving ATSU (R-ATSU) could verify, independently of LOF/NAN exchanges, whether it has been correctly notified to the aircraft as CPDLC NDA. This could avoid the R-ATSU trying to establish CPDLC too early or unnecessarily. Indeed, if an expected OLDI NAN message was not received by the R-ATSU in due time, the R-ATSU can check with the Logon Service whether the T-ATSU notified that R-ATSU is the NDA. Therefore, even in the absence of the NAN message, the R-ATSU can perform CPDLC connection establishment in full confidence.
- the Logon Service could keep track of a flight's current CDA, supporting future implementations of the 4DTRAD service (in which CPDLC clearances originating from downstream ATSUs have to be delivered to an aircraft through the CPDLC CDA-ATSU).

Other benefits in making the Logon Service aware of the CPDLC CDA and NDA is outside the scope of this CONOPS document.

6.3 Network Manager Procedures

As described in the "Network 4DT CONOPS" [Ref 10], the Network Manager will perform the following actions on the receipt of ADS-C/EPP reports from the ADS-C common service. There is no impact on the content of this CONOPS document.

The Network 4DT CONOPS identifies operational constraints in using the data:

"However, EPP cannot be used as a replacement of NM system trajectories, as EPP will not contain many elements provided to Network Manager in post departure period. The EPP elements that might be used to update the NM trajectory relate to the EPP portion of trajectory cleared by ATC, as the remaining part of EPP is not fully stabilised and downstream constraints known to NM may still not be part of FMS trajectory."

And the procedures that will apply once the trajectory information is updated:

"For EPP equipped aircraft, in the case of ATC clearances that trigger a revision of the FMS trajectory and the downlink of a new EPP, this new EPP will be distributed to the network manager through the ADS-C common service. If appropriate, the Network manager will use a portion of this information to update the Network 4D trajectory. The revised trajectory will be distributed again to the subscribed users, in this case the downstream ANSPs."

For ANSP that do not intend to use the NM trajectory services, the trajectory modification will be distributed via APL/ACH mechanism and/or their NMB2B equivalents."

In addition to the Network Manager being a user of the ACS to enhance trajectory information, the Network Manager would benefit in accessing the Logon Service to facilitate its datalink monitoring tasks. Access to the Logon Service is therefore foreseen in section 3.3.

6.4 ADS-C Common Service

6.4.1 End of Flight/Contract Procedures

At the end of flight or when the aircraft leaves the airspace covered by the ACS, the ACS will terminate any existing contracts.

The access to flight status information and/or aircraft position can be used to detect and validate the end of flight or when aircraft leave the covered airspace. In addition, after a certain inactivity period the flight is considered complete and the flight will be removed from ACS active data after a pre-defined period (e.g., 12 hours).

Once the flight is considered complete and the flight is removed from ACS active data, all data relative to the flight shall be stored by the ACS for post-operational analysis and accident/incident investigations. This storage requirement covers all data exchanges with the aircraft (including CM), and data exchanges with ADS-C users that made use of the data.

It is to be noted that the processing of ADS-C/EPP data by third party ADS-C users is not controlled by the ACS, therefore specific clauses must be part of the NDA or licence arrangements between the ACS provider and those ADS-C users.

6.4.2 Procedures in Case of Detected Aborts

In section 4.4.1 a service characteristic of the ACS has been identified as *"The ACS will attempt to recover from any aborted ADS-C dialogues considering any known capabilities of the concerned aircraft or reported avionics issues (such as those experienced during SESAR PJ38 demonstrator)"* and under contract management in section 4.4.2.2.1 *"In the event of notified provider abort, the ACS will attempt to re-establish prior contracts."*

At a high level this can be implemented by applying the following procedures which will be further detailed in the service specifications:

- In case of provider aborts the ACS will try to re-establish prior contracts. This is done by sending after a time interval new ADS-C contract requests, until re-establishment is successful, or end of the flight is detected.
- A user abort with `reason=undefined` is the nominal indication of "end of flight", the ACS will not try to re-establish.
- A user abort with `reason=undefined` while the aircraft is still at the departure gate may occur, and the ACS will attempt to re-establish the connection before departure. This is to mitigate experienced user aborted ADS-C contracts 2min after they were established while engines were still off, and the cabin doors were still open (unintended avionics behaviour). By considering the aircraft position, flight status or any received ADS-C/EPP data, the ACS will attempt to re-establish such user aborted connections at appropriate intervals before departure.
- User aborts triggered by the flight crew while the aircraft is airborne (user abort with `reason=undefined`): For implementations allowing the flight crew to manually terminate all existing ADS-C contracts, procedures and conditions need to be established defining the pre-requisites to re-establish the contract. There might be various reasons for a manual termination e.g., implausible trajectories. Once any issues have been resolved, the flight crew will consent to a new contract activation by performing a new logon to the Logon Service. Alternatively, where implemented, the flight crew can manually disable the ADS-C application (e.g., ADS SET-OFF menu option) thereby terminating all contracts with all ATSUs and the ACS.
- User aborts for other reasons (i.e.: unknown- integrity-check, validation failure, unable-to-decode-message) indicate end to end system misbehaviour. ACS will not try to re-establish in such cases.

6.4.3 Procedure to Increase Logon Connection Rates

The Logon Service depends on the initial manual logon by the flight crew, yet current datalink service monitoring reveals a low logon connection rate by crews (44% on average). As a result,

it can be assumed that the downlinking of ADS-C/EPP data will be negatively impacted by low logon connection rates and mitigation should be foreseen.

The ACS can derive aircraft position data from NM B2B Flight Services (or even ADS-B) and flight status data from NM B2B Flight Services. Therefore, the ACS can determine a time by which the crew should have logged on and by which addressing information should have been available from the Logon Service.

In the event the expected addressing information is missing, the ACS can request the Logon Service to uplink a CM-contact to the aircraft, thereby forcing the aircraft to issue a CM-logon request to the Logon Service to obtain the information. Depending on the aircraft implementation, receipt of a CM-contact prior to any CM-logon may be rejected as this does not comply with ICAO Doc 9880 Part 1 clause 2.3.5.1.1. However, this procedure could ensure access to addressing information so that ADS-C contracts can be established after successful DLIC initiation (ICAO Doc 9880 and EUROCAE ED-228).

This procedure requires access to aircraft position, flight status and a database of ATS B2 ADS-C capable aircraft with corresponding TSAP addresses. Although it does not incur additional safety concerns, the ACS will need to be designed to handle the following two cases:

- An airline operator may use a different aircraft from the one indicated in FPL – the CM-contact will fail or sent to the wrong aircraft. The ACS will need to check the flight ARCID/ADEP/ADES to ensure the CM-logon request is not coming from the wrong aircraft
- The known aircraft TSAP address could be wrong or out of date. In this case, there will be no answer to CM-contact from the aircraft

Even if ANSPs receive/request the logon information from the Logon Service following this CM-contact CM-logon sequence, CPDLC avionics will remain in the inhibited state as there was no crew-initiated logon – CPDLC cannot be established. This procedure has been tested by SESAR PJ38 (October 2022) with a limited number of participating aircraft. The number of logons increased three-fold, however, when the procedure was unsuccessful subsequent uplinks of CM-contact requests at 10-minute intervals became less effective.

This procedure will be an option for the Logon and ADS-C Common Services with guidance on its use in particular with regard to the number and frequency of uplinked CM-contact requests to trigger aircraft CM-logon requests. Essentially, if logon rates do not pick up with improved and defragmented CPDLC services, ADS-C/EPP benefits for AUs and the establishment of the Logon Service (as a means to tentatively improve logon rates), this procedure will be brought forward. Therefore, it needs to be a recommended requirement for the service specifications.

6.5 Logon Service and ACS

The service availability of the Logon Service and the ACS must be known to ANSPs, NM and other ADS-users. Procedures for Logon Service and ACS in the event of failures and aborts will need to be documented in the service specifications to be developed by OEP Workstream 12.2 (DEL2.1-4 of the topic plan [Ref 1] on common logon and ADS-C services). These may be also supported by the guidelines (DEL2.3-1 of the topic plan [Ref 1]).

Annex A. Initial Options for the Logon Service

A.1. Introduction

This Annex documents the initial 12 options that were identified **to enable a single CM-logon request by the flight deck crew**. The options have been assessed and evaluated by the workstream topic members leading to the definition of the Logon Service in section 3 of this CONOPS. This Annex provides insight to the issues that were addressed, scoring methodologies and the rationale of their combination.

Most of the options focus on how to ground distribute CM-logon information between ANSPs, the Logon Service and the ACS. Some options are air-ground driven by automating the CM-logon or 'forcing' CM-logons because of ground uplinked CM-contact requests.

In line with the colour scheme of section 2, Figure 6 illustrates the single CM-logon request containing B1 and B2 information addressed to the first ACC or the Logon Service. Then the Logon Service and ANSPs depend on each other to exchange received logon information. The flowchart indicates the series of steps leading to the identified options (1 to 9).

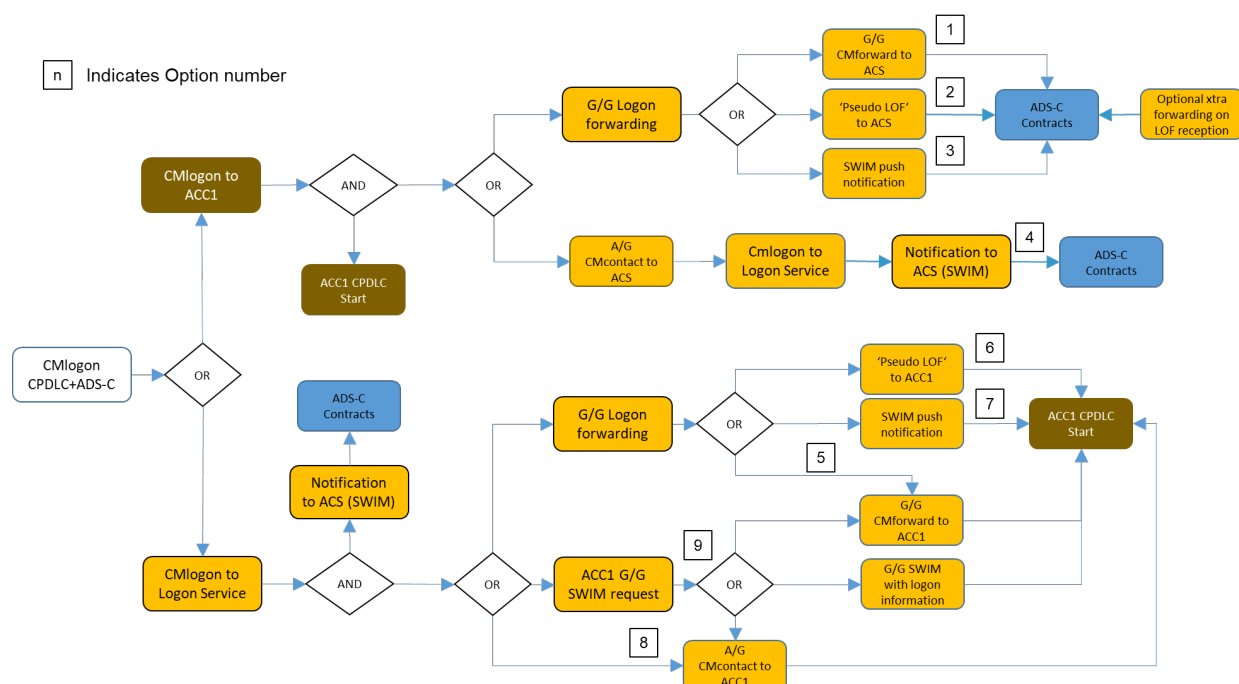


Figure 6: Single CM-logon by Flight Deck Crew (ACCs and Logon Service Interdependent)

Figure 7 identifies an additional set of options (10 to 12) have also been considered whereby the flight deck initiates a single CM-logon request to ANSPs but other means are deployed to enable the ADS-C application service. The Logon Service and/or the ACS operate independently of ANSPs.

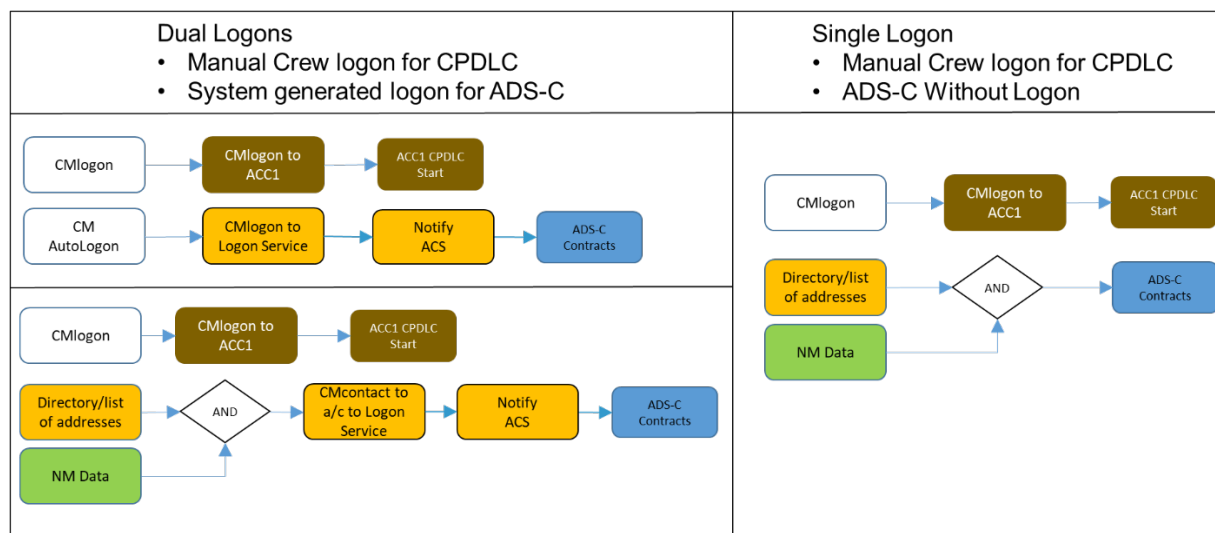


Figure 7: Single CM-logon by Flight Deck Crew (ACCs and Logon Service Independent)

A.2. Description of the Options

Each option describes a 'nominal' procedure to invoke the CM-logon and sharing of logon information between ACCs, the Logon Service and the ACS. A basic summary of the data flows, a diagram and a table indicating the identified pros/cons and impacts are provided. It is to be noted that in each option 'ACC1' refers to first ACC that receives the initial CM-logon from the aircraft.

In the below options, the use of thick orange arrows with red outline indicates new data exchanges that are either already standardised but have never been deployed or need to be standardised. For convenience, the figures illustrating each option co-locate the Logon Service and the ACS.

A.2.1. [Option 1] CM-logon to ANSP (ACC1) and CM-forward to Logon Service

Description The CM-logon request containing B1 and B2 logon information is sent to the first ACC on the flight trajectory. Upon receipt of a CM-logon or a LOF message, a ground-ground ICAO standard CM-forward message is sent to transmit the logon information from ACCs to the Logon Service over the current ATN network. ACCs will receive EPP data from the ACS over the SWIM interface using NewPENS.

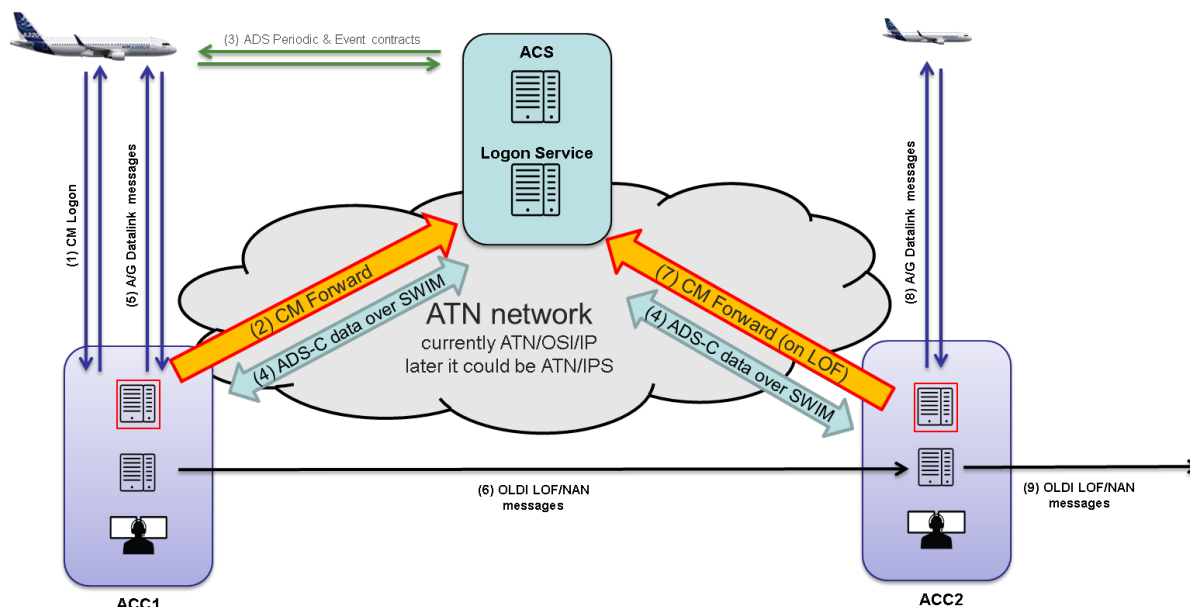


Figure 8: [Option 1] CM-logon Request to ACC and CM-forward

Pros	Cons	Impact Assessment
1. Same flight deck logon procedures applied for B2 as for current B1 (no changes) 2. Makes use of existing standard for CM-forward (ICAO 9880 [Ref 3]) 3. Minimal changes to ANSPs to support CM-forward request (CM-forward ASN.1 identical to CM-logon) 4. Once CM-forward is implemented on a commonly developed ATC front end system (e.g., DLFEP), it becomes available to several ANSPs 5. CM-forward is duplicated on LOF receipt (redundancy) 6. Logon Service can store all address information – a data broker or backup database of addresses	1. All ACCs need to support CM-logon with B1+B2 2. Requires modification of all ATC end systems (datalink FEPs) to support CM-forward 3. Impact of delayed CM-logon to ACCs on lower centre access to EPP data (e.g., 1st logon may only occur when FPL is activated by ACC) 4. For most ACCs, CM-logon data is lost if there is no matching FPL 5. ACCs depend on the ACS to establish ADS-C contracts before receiving EPP data 6. Underlying ground IP infrastructure to be agreed (i.e., NewPENS) if different to current ATN/OSI network to transmit CM-forward	ANSP: 1. Application based logon list (i.e., adapted behaviour for CPDLC and ADS-C) 2. Ensure unack'd LOF does not inhibit CM-forward to the Logon Service 3. Implement CM-forward Request, even if CM-logon is rejected by Logon List 4. All centres need to support ATS B2 CM (CRO-1159, CRO-1212, CRO-1213, CRO-1214) ACS and Logon Service: 1. ACS is SWIM client of Logon Service 2. Needs to comply to CM-server requirements 3. Filter or overwrite CM-forward duplicates 4. Define a means to retrieve stored CM logon address information Airspace Users: 1. No impact on current B1/B2 fleet Regulations and Standards: 1. Note that ED229 §3.1.4.5.1 on CM-forward relates to LOF exchanges and is not to be applied for ACC exchanges with the Logon Service

A.2.2. [Option 2] CM-logon to ANSP (ACC1) and 'Pseudo-LOF' to Logon Service

Description The CM-logon request containing B1 and B2 logon information is sent to the first ACC on the flight trajectory. Upon receipt of a CM-logon or a LOF message, a **new** European defined ground-ground message (pseudo-LOF in ASN.1 or XML) is sent to distribute the logon information from ACCs to the Logon Service over NewPENS. The term 'pseudo' is to indicate that it would be like (or an extension of) the OLDI LOF message, yet of different purpose as it not exchanged between two ACCs. ACCs will receive EPP data from the ACS over the SWIM interface.

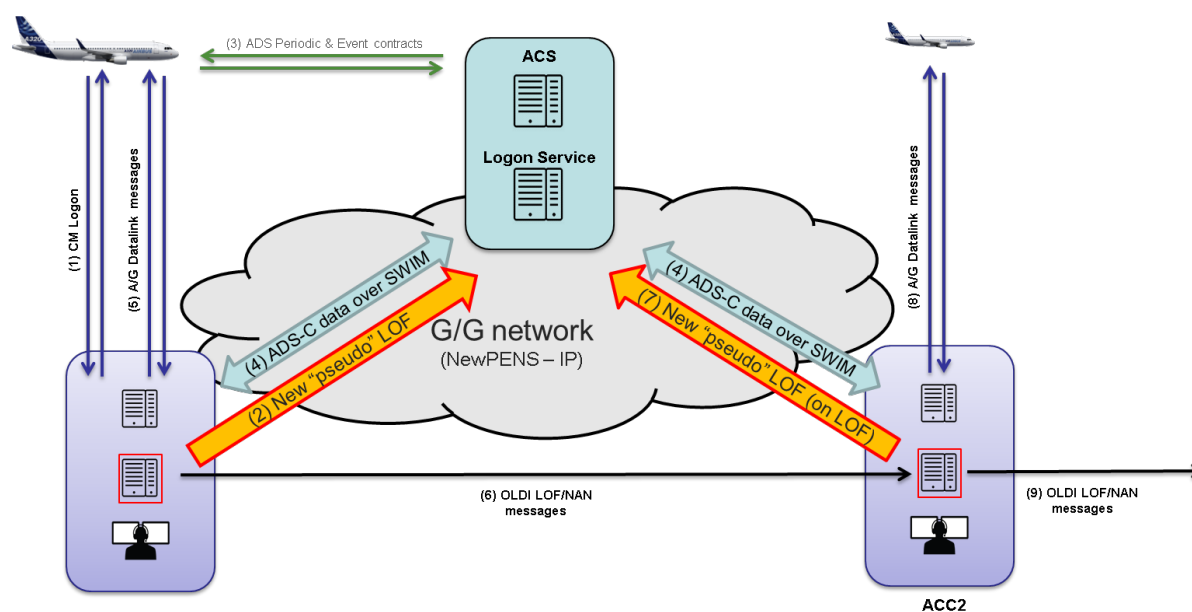


Figure 9: [Option 2] CM-logon Request to ACC and Pseudo-LOF

Pros	Cons	Impact Assessment
- Same flight deck logon procedures applied for B2 as for current B1 (no changes) - Any new pseudo-LOF functionality that is implemented on a commonly developed ATC front end system (e.g., DLFEF), becomes available to several ANSPs 'pseudo-LOF' is duplicated on LOF receipt (redundancy) - Opportunity to define a new future proof format based on IP ('pseudo-LOF') - Logon Service can store all address information – a data broker or backup database of addresses	- All ACCs need to support CM-logon with B1+B2 - Need to agree on 'pseudo-LOF' message format and its link to operational events - Requires modification of all ground systems (ACC) to support 'pseudo-LOF' - Impact of delayed CM-logon to ACCs on lower centre access to EPP data (e.g., 1st logon may only occur when FPL is activated by ACC) - For most ACCs, CM-logon data is lost if there is no matching FPL - ACCs depend on ACS to establish ADS-C contracts before receiving EPP data - Underlying ground IP infrastructure to be agreed (i.e. NewPENS) to transmit pseudo-LOF	ANSP: - Application based logon list (i.e., adapted behaviour for CPDLC and ADS-C) - Ensure unack'd LOF does not inhibit 'pseudo-LOF' forwarding to the Logon Service - Implement 'pseudo-LOF', even if CM-logon is rejected by Logon List - All centres need to support ATS B2 CM (CRO-1159, CRO-1212, CRO-1213, CRO-1214) ACS and Logon Service: - Filter or overwrite 'pseudo-LOF' duplicates - Define a means to retrieve stored logon address information Airspace Users: - No impact on current B1/B2 fleet Regulations and Standards: 'Pseudo-LOF' to be standardised as part of the OEP WST 12.2 specifications

A.2.3. [Option 3] CM-logon to ANSP (ACC1) and SWIM Distribution to Logon Service

Description The CM-logon request containing B1 and B2 logon information is sent to the first ACC on the flight trajectory. Upon receipt of a CM-logon or a LOF message, a new additional SWIM service is used to distribute the logon information from ACCs to the Logon Service over NewPENS. ACCs could receive EPP data from the ACS over the same SWIM interface.

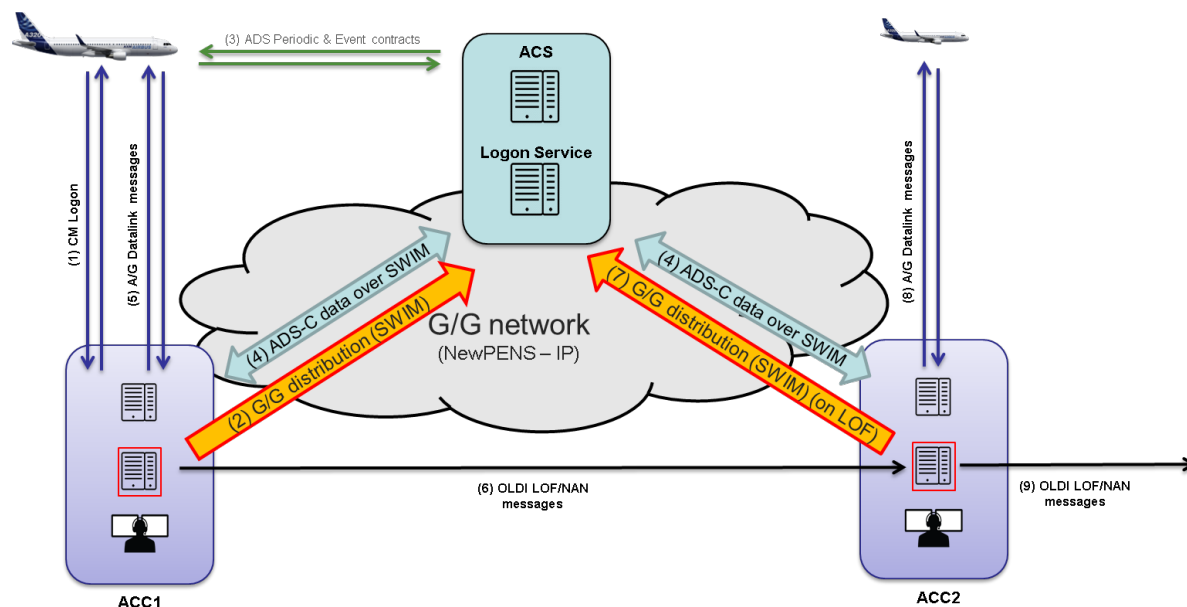


Figure 10: [Option 3] CM-logon Request to ACC and SWIM

Pros	Cons	Impact Assessment
1. Same flight deck logon procedures applied for B2 as for current B1 (no changes) 2. Any new SWIM functionality that is implemented on a commonly developed ATC front end system (e.g., DLFEP), becomes available to several ANSPs 3. Updates via SWIM on LOF receipt (redundancy) 4. Future proof by using SWIM for ground distribution 5. Logon Service can store all address information – a data broker or backup database of addresses 6. Implementation can be combined with the SWIM development to receive ADS-C/EPP data	1. All ACCs need to support CM-logon with B1+B2 2. Impact of delayed CM-logon to ACCs on lower centre access to EPP data (e.g., 1st logon may only occur when FPL is activated by ACC) 3. For most ACCs, CM-logon data is lost if there is no matching FPL 4. ACCs depend on ACS to establish ADS-C contracts before receiving EPP data 5. Underlying ground IP infrastructure to be agreed (i.e., NewPENS) to transmit SWIM data exchanges of logon information	ANSP: 1. Can require substantial modification of all ground systems (ACC) to support new SWIM interface if separate from ADS/EPP interface 2. Application based logon list (i.e., adapted behaviour for CPDLC and ADS-C) 3. Ensure unack'd LOF does not inhibit SWIM update to the Logon Service 4. Implement SWIM update, even if CM-logon is rejected by Logon List 5. All centres need to support ATS B2 CM (CRO-1159, CRO-1212, CRO-1213, CRO-1214) ACS and Logon Service: 1. Add new SWIM interface definition 2. Define a means to retrieve stored logon address information 3. Filter or overwrite SWIM updates Airspace Users: 1. No impact on current B1/B2 fleet Regulations and Standards: 1. SWIM service additions to be standardised as part of the OEP WST 12.2 specifications (logon data exchanges)

A.2.4. [Option 4] CM-logon to ANSP (ACC1) and CM-contact to Logon Service

Description The CM-logon request containing B1 and B2 logon information is sent to the first ACC on the flight trajectory. Upon receipt of the CM-logon request, ACC1 uplinks a CM-contact so that the aircraft sends a second CM-logon to the Logon Service. On receipt of a LOF message, ACCs can uplink a CM-contact so that the aircraft issues a CM-logon to the Logon Service to ensure the availability of logon information. ACCs receive EPP data from the ACS over the SWIM interface.

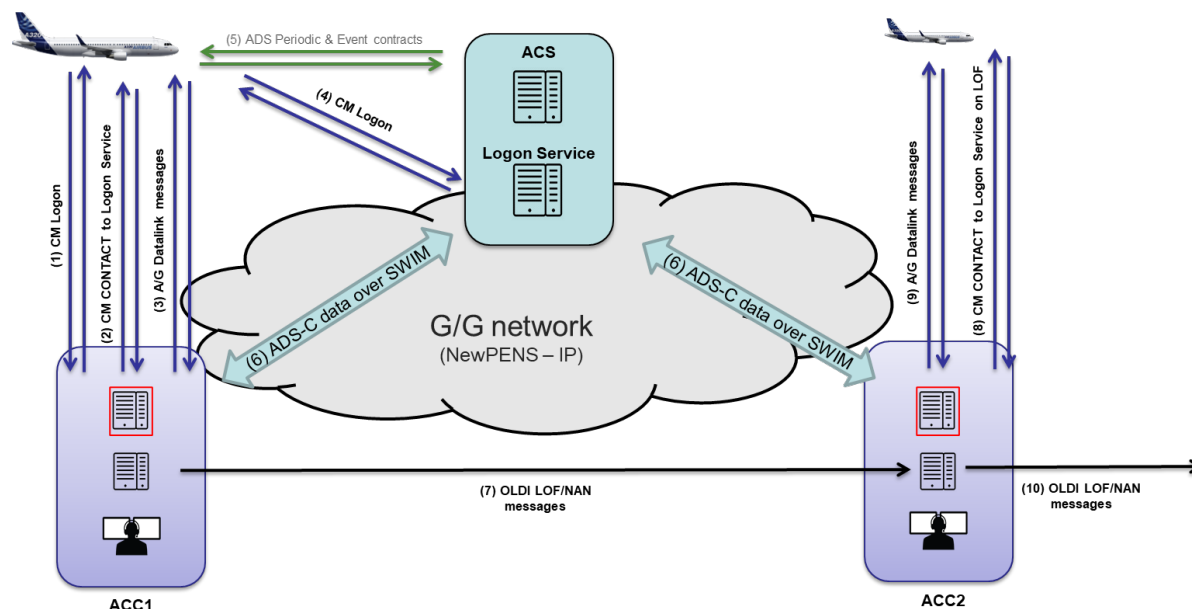


Figure 11: [Option 4] CM-logon Request to ACC and CM-contact to Logon Service

Pros	Cons	Impact Assessment
1. Same flight deck logon procedures applied for B2 as for current B1 (no changes) 2. Uses existing A/G messages (CM-contact) 3. Any new CM-contact logic that is implemented on a commonly developed ATC front end system (e.g., DLFEF), becomes available to several ANSPs 4. Logon Service can store all address information – a data broker or backup database of addresses	1. All ACCs need to support CM-logon with B1+B2 2. Impact of delayed CM-logon to ACCs on lower centre access to EPP data (e.g., 1st logon may only occur when FPL is activated by ACC) 3. For most ACCs, CM-logon data is lost if there is no matching FPL 4. ACCs depend on ACS to establish ADS-C contracts before receiving EPP data 5. ACC2 does not know whether the aircraft successfully issued a CM-logon to the Logon Service which may require to uplink CM-contact on LOF reception	ANSP: 1. Implement CM-contact logic on ATC front end system and/or FDPS depending on local architecture (at least for B2 fleet) 2. Issue new CM-contact on LOF reception in case aircraft fails to logon to the Logon Service 3. All centres need to support ATS B2 CM (CRO-1159, CRO-1212, CRO-1213, CRO-1214) ACS and Logon Service: 1. ACS is SWIM client of Logon Service 2. Needs to comply to CM-server requirements 3. Define a means to retrieve stored CM logon address information Airspace Users: 1. No impact on airborne ATC database as the GFD and NSAP address of the Logon Service would be indicated in the uplinked CM-contact

		Regulations and Standards: Note that ED229 §3.1.4.5.1 on CM-forward relates to LOF exchanges and is not to be applied for ACC exchanges with Logon Service
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A.2.5. [Option 5] CM-logon to Logon Service and CM-forward to ANSP (ACC1)

Description The CM-logon request containing B1 and B2 logon information is sent to the Logon Service. Upon receipt of the CM-logon and available NM flight status and positional data, a ground-ground ICAO standard CM-forward message is sent to distribute the logon information from the Logon Service to the appropriate ACC(s) over the current ATN network. ACCs will receive EPP data from the ACS over the SWIM interface using NewPENS.

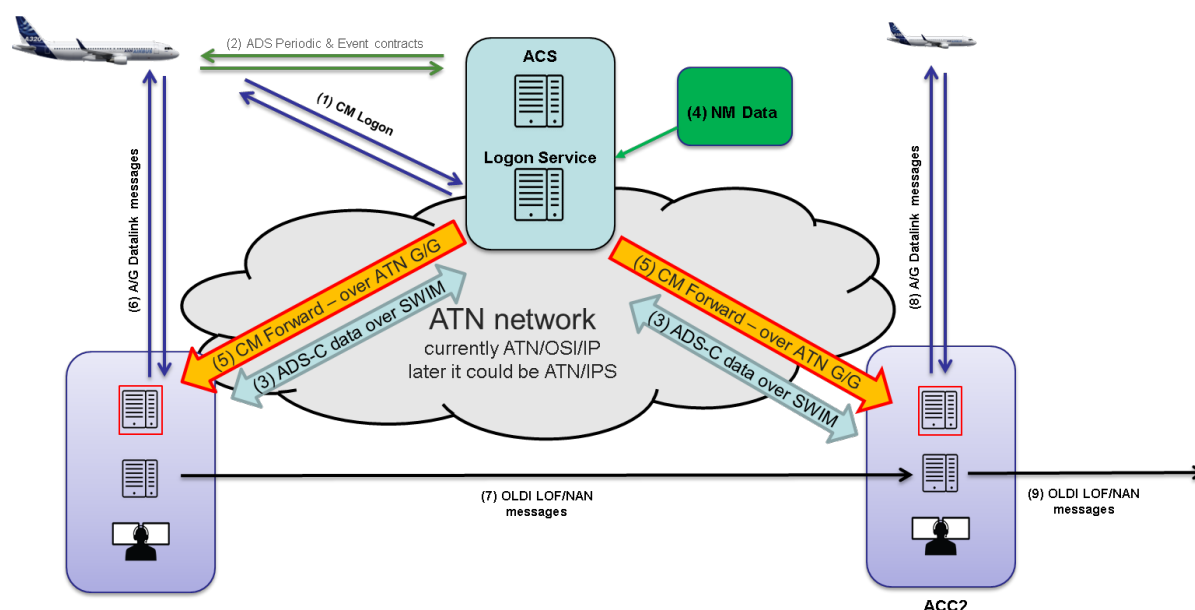


Figure 12: [Option 5] CM-logon Request to Logon Service and CM-forward

Pros	Cons	Impact Assessment
- Single address for flight crews to perform manual logon (simpler procedure) - Makes use of existing standard for CM-forward (ICAO 9880 [Ref 3]) - Minimal changes to ANSPs to support CM-forward reception (CM-forward ASN.1 identical to CM-logon) - Once CM-forward is implemented on a commonly developed ATC front end system (e.g., DLFEP), it becomes available to several ANSPs - Logon Service can store all address information – a data broker or backup database of addresses - ADS-C/EPP immediately available to all users (no	- Change in procedure for crews to logon to the Logon Service - Initial ACC (ACC1) dependency on Logon Service forwarding for CPDLC connections - Logon Service may target the 'CDA' too late or the 'NDA' too early (to start CPDLC – leading to failure) - Logon Service unawareness in cases of re-routing between ANSPs may affect identification of ACC1 and mis-direct the CM-forward - All ACCs will still need to support CM-logon for legacy B1 (or mis-directed B2 CM-logons) - Underlying ground IP infrastructure to be agreed (i.e., NewPENS) if different to	ANSP: - Implement CM-forward reception (incl. B1+B2 data) to perform CPDLC - Requires limited modification of all ground systems (ACC) to change CM-logon processing (e.g., DLFEP to send the CM-forward data to FDPS) - Likely loss of CM-forward logon information if ANSP is not ready (e.g., inactive flight plan) - ACCs will still need to forward any received CM-logon information to the Logon Service using CM-forward ACS and Logon Service: - ACS is SWIM client of Logon Service - Needs to comply to CM-server requirements - Needs to increase its SWAL level

dependency on 1st CM-logon to ACC) 7. Evolution from data chain forwarding (e.g., LOF) to data distribution from common service	current ATN/OSI network to transmit CM-forward	4. Define a means to retrieve stored CM logon address information 5. Distribute CM-logon information when A/C is close to AoR of each ACC (tracking function or NM data) 6. CM-forward multicast or broadcast to avoid mis-directs Airspace Users: 1. Add a GFD for existing fleet to enable CM-logon to Logon Service Regulations and Standards: 1. Note that ED229 §3.1.4.5.1 on CM-forward relates to LOF exchanges and is not to be applied for ACC exchanges with Logon Service
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A.2.6. [Option 6] CM-logon to Logon Service and 'Pseudo-LOF' to ANSP (ACC1)

Description The CM-logon request containing B1 and B2 logon information is sent to the Logon Service. Upon receipt of the CM-logon and available NM flight status and positional data, a **new** European defined ground-ground message (pseudo-LOF in ASN.1 or XML) is sent to distribute the logon information from the Logon Service to the appropriate ACC(s) over NewPENS. The term 'pseudo' is to indicate that it would be like (or an extension of) the OLDI LOF message, yet of different purpose as it not exchanged between two ACCs. ACCs will receive EPP data from the ACS over the SWIM interface.

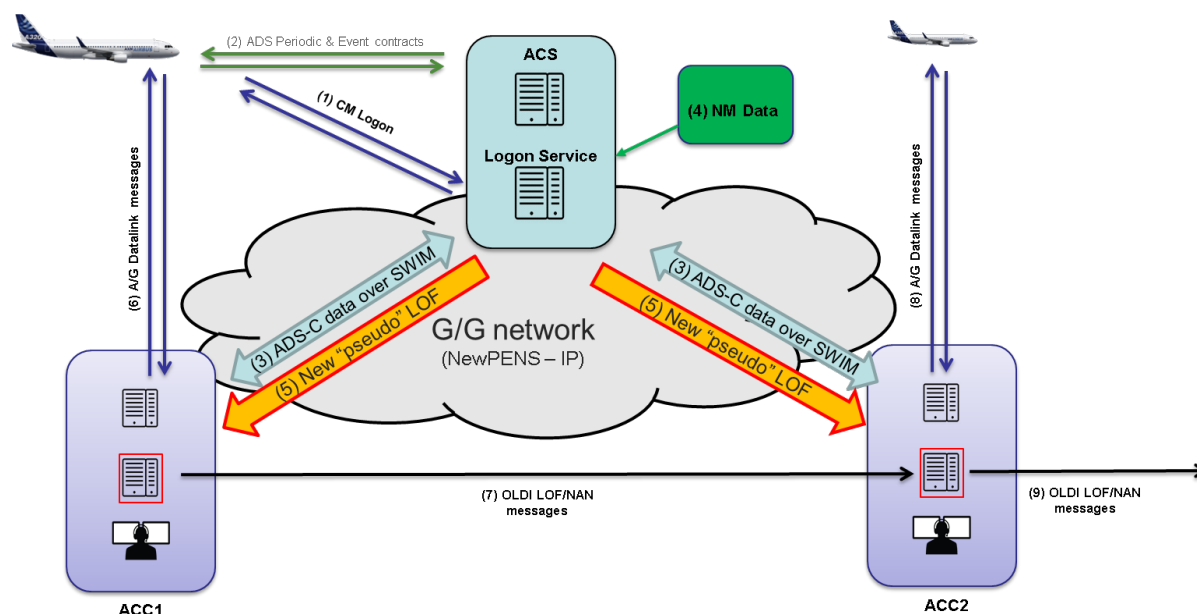


Figure 13: [Option 6] CM-logon Request to Logon Service and Pseudo-LOF to ACC

Pros	Cons	Impact Assessment
1. Single address for flight crews to perform manual logon (simple procedure) 2. Any new pseudo-LOF functionality that is implemented on a commonly developed ATC front end system (e.g., DLFEF), becomes available to several ANSPs	1. Change in procedure for crews to logon to the Logon Service 2. Need to agree on 'pseudo-LOF' message format and its link to operational events 3. Requires modification of all ground systems (ACC) to support 'pseudo-LOF'	ANSP: 1. Implement pseudo-LOF reception (incl. B1+B2 data) to perform CPDLC 2. Requires processing of incoming pseudo-LOF by ATC front end system and/or FDPs depending on local ACC architecture

- Logon Service can store all address information – a data broker or backup database of addresses - ADS-C/EPP immediately available to all users (no dependency on 1st CM-logon to ACC) - Evolution from data chain forwarding (e.g., LOF) to data distribution from common service	- Initial ACC (ACC1) dependency on Logon Service forwarding for CPDLC connections - Logon Service may target the 'CDA' too late or the 'NDA' too early (to start CPDLC – leading to failure) - Logon Service unawareness in cases of re-routing between ANSPs may affect identification of ACC1 and mis-direct the pseudo-LOF - All ACCs will still need to support CM-logon for legacy B1 (or mis-directed B2 CM-logons) - Underlying ground IP infrastructure to be agreed (i.e., NewPENS) to transmit pseudo-LOF	- Likely loss of pseudo-LOF logon information if ANSP is not ready (e.g., inactive flight plan) - ACCs will still need to forward any received CM-logon information to the Logon Service using pseudo-LOF ACS and Logon Service: - ACS is SWIM client of Logon Service - Needs to comply to CM-server requirements - Needs to increase its SWAL level - Define a means to retrieve stored CM logon address information - Distribute CM-logon information when A/C is close to AoR of each ACC (tracking function or NM data) - Pseudo-LOF multicast or broadcast to avoid mis-directs Airspace Users: - Add a GFD for existing fleet to enable CM-logon to Logon Service Regulations and Standards: 'Pseudo-LOF' to be standardised as part of the OEP WST 12.2 specifications
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A.2.7. [Option 7] CM-logon to Logon Service and SWIM Distribution to ANSP (ACC1)

Description The CM-logon request containing B1 and B2 logon information is sent to the Logon Service. Upon receipt of the CM-logon and available NM flight status and positional data, a new additional SWIM service is used to distribute the logon information from the Logon Service to the appropriate ACC(s) over NewPENS. ACCs could receive EPP data from the ACS over the same SWIM interface.

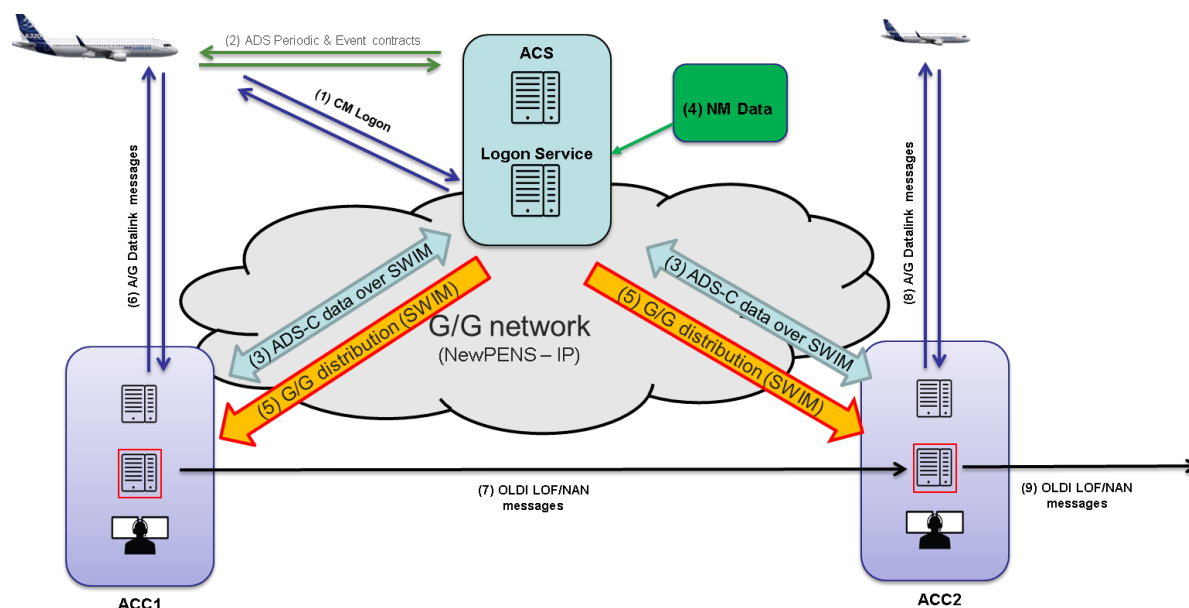


Figure 14: [Option 7] CM-logon Request to Logon Service and SWIM

Pros	Cons	Impact Assessment
1. Single address for flight crews to perform manual logon (simple procedure) 2. Future proof by using SWIM for ground distribution 3. Implementation can be combined with the SWIM development to receive ADS-C/EPP data 4. Any new SWIM functionality that is implemented on a commonly developed ATC front end system (e.g., DLFEP), becomes available to several ANSPs 5. Logon Service can store all address information – a data broker or backup database of addresses 6. ADS-C/EPP immediately available to all users (no dependency on 1st CM-logon to ACC) 7. Evolution from data chain forwarding (e.g., LOF) to data distribution from common service	1. Change in procedure for crews to logon to the Logon Service 2. SWIM service to be standardised 3. Can require substantial modification of all ground systems (ACC) to support new SWIM interface if separate from ADS/EPP interface 4. Initial ACC (ACC1) dependency on Logon Service forwarding for CPDLC connections 5. Logon Service may target the 'CDA' too late or the 'NDA' too early (to start CPDLC – leading to failure) 6. Logon Service unawareness of re-routing between ANSPs may affect identification of ACC1 and mis-direct the SWIM data exchange 7. All ACCs will still need to support CM-logon for legacy B1 (or mis-directed B2 CM-logons) 8. Underlying ground IP infrastructure to be agreed (i.e., NewPENS) to transmit SWIM data exchanges of logon information	ANSP: 1. Requires processing of incoming SWIM data by ATC front end system or FDPS) depending on local ACC architecture 2. Likely loss of SWIM logon information if ANSP is not ready (e.g., inactive flight plan) 3. ACCs will still need to forward any received CM-logon information to the Logon Service using SWIM ACS and Logon Service: 1. ACS is SWIM client of logon Service 2. Needs to comply to CM-server requirements 3. Needs to increase its SWAL level 4. Add new SWIM interface definition 5. Define a means to retrieve stored CM logon address information 6. Distribute CM-logon information when A/C is close to AoR of each ACC (tracking function or NM data) 7. SWIM multicast, broadcast, or publication to avoid mis-directs Airspace Users: 1. Add a GFD for existing fleet to enable CM-logon to Logon Service Regulations and Standards: 1. SWIM service additions to be standardised as part of the OEP WST 12.2 specifications (logon data exchanges)

A.2.8. [Option 8] CM-logon to Logon Service and CM-contact to ANSP (ACC1)

Description The CM-logon request containing B1 and B2 logon information is sent to the Logon Service. Upon receipt of the CM-logon, the Logon Service uplinks a CM-contact so that the aircraft sends a second CM-logon to the first ACC1. ACCs receive EPP data from the ACS over the SWIM interface using NewPENS.

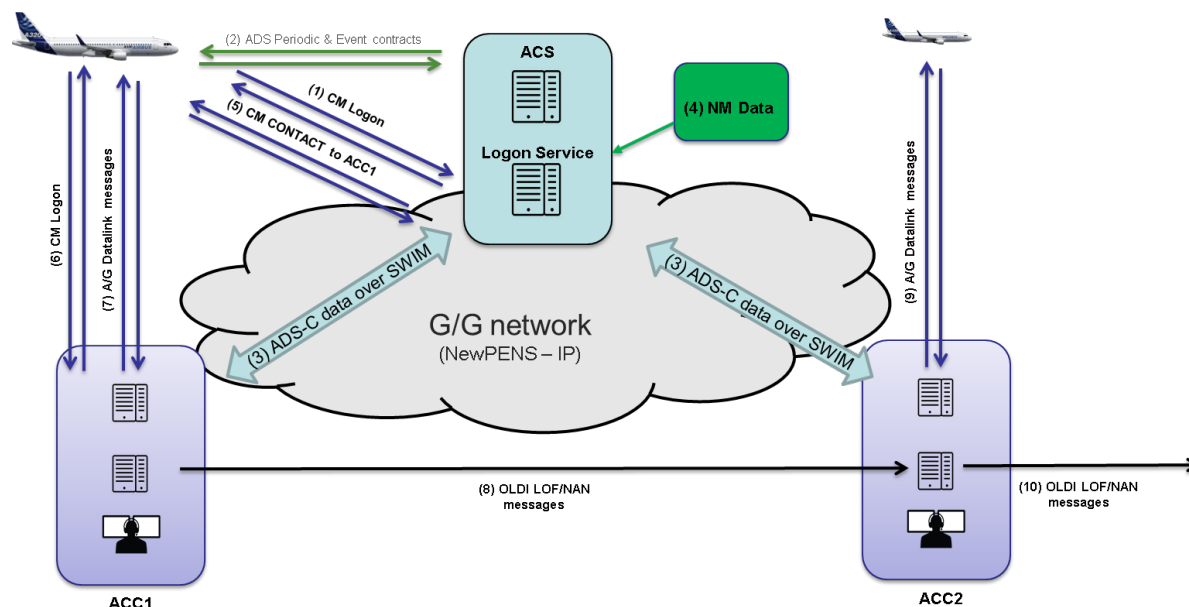


Figure 15: [Option 8] CM-logon Request to Logon Service and CM-contact to ACC

Pros	Cons	Impact Assessment
1. Single address for flight crews to perform manual logon (simple procedure) 2. Using existing A/G messages 3. Less complex NSAP database on board (as they are sent in CM-contact) 4. ADS-C/EPP data is directly available to all 5. Logon Service can store all address information – a data broker or backup database of addresses	1. Change in procedure for crews to logon to the Logon Service 2. All ACCs need to support CM-logon with B1+B2 3. Initial ACC (ACC1) dependency on Logon Service forwarding for CPDLC connections 4. Logon Service may target the 'CDA' too late or the 'NDA' too early (to start CPDLC – leading to failure) 5. ACC1 doesn't know if the CM-logon came via Logon Service or directly from the flight deck, unclear if the Logon Service has the logon information for the ACS.	ANSP: 1. All centres need to support ATS B2 CM (CRO-1159, CRO-1212, CRO-1213, CRO-1214) 2. Possible loss of CM-logon information if ACS triggers CM-contact too early (e.g., ACC1 inactive flight plan) ACS and Logon Service: 1. ACS is SWIM client of Logon Service 2. Needs to comply to CM-server requirements 3. Needs to increase its SWAL level 4. Coupling with NM (or EPP data) to determine ACC1 5. Define a means to retrieve stored CM logon address information Airspace Users: 1. Add a GFD for existing fleet to enable CM-logon to Logon Service Regulations and Standards: N/A

A.2.9. [Option 9] CM-logon to Logon Service and ANSP (ACC1) Request

Description The CM-logon request containing B1 and B2 logon information is sent to the Logon Service. Based on local triggering conditions (e.g., OLDI messages or assume), ACC1 will issue a SWIM request to the Logon Service to receive the logon information. Depending on ACC1 requested response method, the Logon Service will return a ground-ground ICAO standard CM-forward message, a SWIM response or will uplink a CM-contact so that the aircraft sends a second CM-logon to ACC1. ACCs will receive EPP data from the ACS over the SWIM interface using NewPENS.

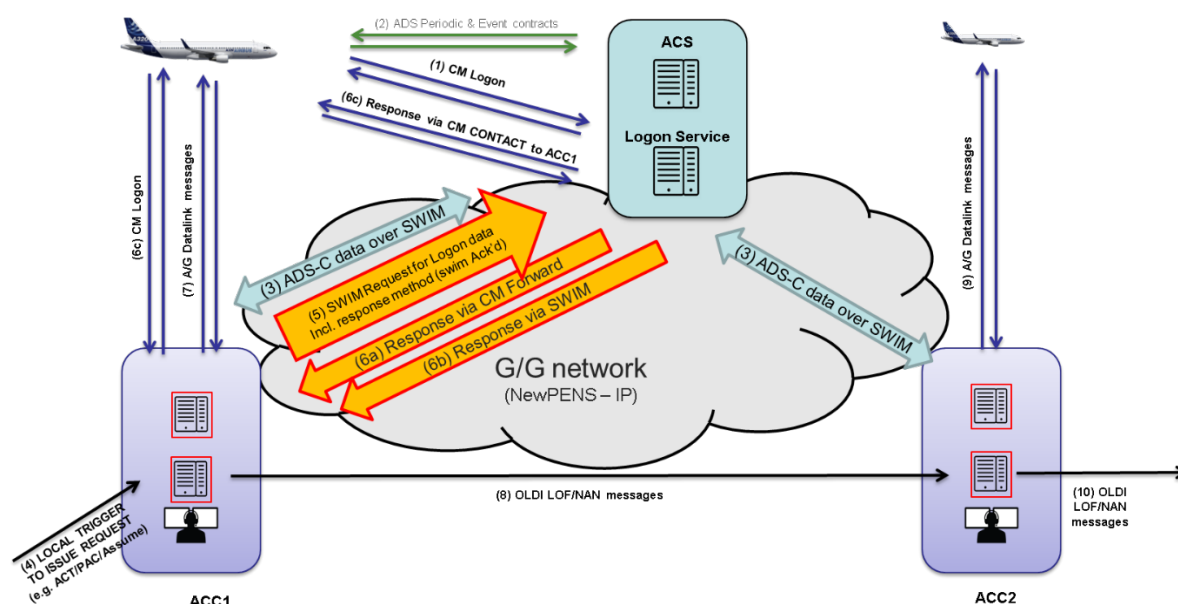


Figure 16: [Option 9] CM-logon Request to Logon Service and ANSP Request

Pros	Cons	Impact Assessment
- Single address for flight crews to perform manual logon (simple procedure) - Makes use of existing standard for CM-forward (ICAO 9880 [Ref 3]) - Minimal changes for ANSPs to support CM-forward reception - Once CM-forward is implemented on a commonly developed ATC front end system (e.g., DLFEF), it becomes available to several ANSP - SWIM interface to receive logon information can be part of the SWIM integration to receive EPP data - Logon Service stores all address information – available to all - ADS-C/EPP immediately available to all users (no dependency on 1st CM-logon to ACC)	- Change in procedure for crews to logon to the Logon Service - All ACCs using CM-contact response method need to support CM-logon with B1+B2 - ACC decides when to issue request depending on its operational context (e.g., ABI, PAC, ACT message or assumed flight) - ACC may request logon information too early requiring an ACC 'subscription' or 'retry' - Request/response via SWIM more complex to implement than CM-forward or CM-contact - All ACCs will still need to support CM-logon for legacy B1 (or mis-directed B2 CM-logons) - Underlying ground IP infrastructure to be agreed (i.e., NewPENS) if different to	ANSP: - All ACCs need to implement CM-forward reception or updated SWIM interface to reduce CM-contact on A/G - Modification of all ground systems (ACC) to request logon information from Logon Service - Define triggering conditions to request to CM logon address information and manage when done too early. - All centres using CM-contact response method need to support ATS B2 CM (CRO-1159, CRO-1212, CRO-1213, CRO-1214) - Request full B1 and B2 address information from Logon Service, if LOF only contains B1 information for B2 aircraft - Timing of request to Logon Service and CPDLC start in cases of missing LOF/NAN from upstream centre ACS and Logon Service:

8. Evolution from data chain forwarding (e.g., LOF) to data distribution from common service 9. ACC2 can also request logon information from the Logon Service if the LOF message contains B1-only information or if LOF chain is broken 10. Based on the principle that each ACC 'knows best' when to request logon information	current ATN network to transmit CM-forward and SWIM data exchanges of logon information	1. ACS is SWIM client of Logon Service 2. Needs to comply to CM-server requirements 3. Needs to increase its SWAL level 4. Needs to implement multiple response options (CM-forward, SWIM, CM-contact) Airspace Users: 1. Add a GFD for existing fleet to enable CM-logon to Logon Service Regulations and Standards: 1. SWIM service additions to be standardised as part of the OEP WST 12.2 specifications (ACC request incl. parameters, logon data access; and case of ACC requesting address information before CM-logon)
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A.2.10. [Option 10] CM-logon to ANSP (ACC1) and Autologon to Logon Service

Description An automatic CM-logon request (on-board system generated) is sent to the Logon Service. Independently, the flight deck crew initiates a second (manual) CM-logon request containing B1 and B2 logon information which is sent to the first ACC on the flight trajectory. ACCs will receive EPP data from the ACS over the SWIM interface using NewPENS.

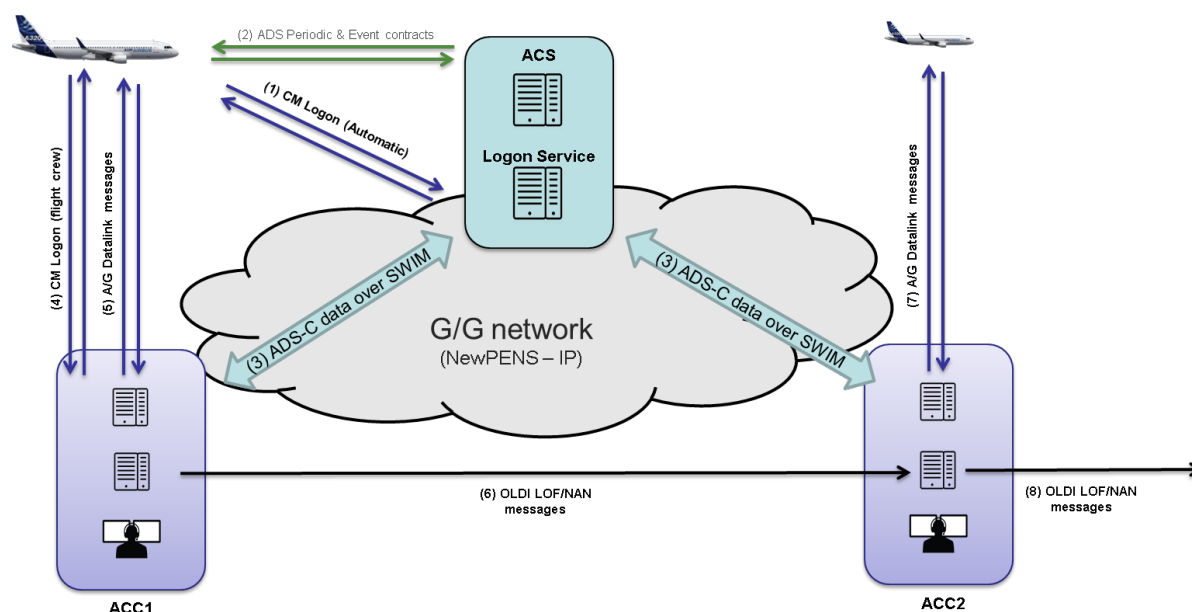


Figure 17: [Option 10] CM-logon Request to ACC1 and AutoLogon to Logon Service

Pros	Cons	Impact Assessment
1. Same flight deck logon procedures applied for B2 as current B1 (no changes) 2. Autologon independent of crew actions,	1. All ACCs need to support CM-logon with B1+B2 2. Not supported by current certified avionics 3. Requires avionic development 4. Affects all B2 avionics	ANSP: 1. All centres need to support ATS B2 CM (CRO-1159, CRO-1212, CRO-1213, CRO-1214)

3. ADS-C/EPP data is available to all after Autologon to Logon Service 4. Logon Service can store all address information – a data broker or backup database of addresses	5. Availability for CP1 AF6? 6. Lack of any awareness between ACC/Logon Service due to two separate CM-logons	ACS and Logon Service: 1. ACS is SWIM client of Logon Service 2. Needs to comply to CM-server requirements 3. Triggering conditions for CM-contact for take-off and incoming flights 4. Implement A/C tracking function using NM data 5. Define a means to retrieve stored CM logon address information Airspace Users: 1. Availability of avionics for forward-fitting 2. Availability of retrofitting options for existing B2 fleet 3. No impact on current B1 fleet which do not need to connect to the Logon Service for the ACS Regulations and Standards: 1. Impact on ICAO/EUROCAE standards unknown (unfeasible for the CP1 industrial target date), proposal can be forwarded to WG78
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A.2.11. [Option 11] CM-logon to ANSP (ACC1) and Logon Service CM-contact

Description The Logon Service issues a CM-contact to the aircraft forcing a CM-logon to itself by relying on available addressing and available NM flight status and positional data. Independently, the flight deck crew initiates a second (manual) CM-logon request containing B1 and B2 logon information which is sent to the first ACC on the flight trajectory. ACCs will receive EPP data from the ACS over the SWIM interface using NewPENS.

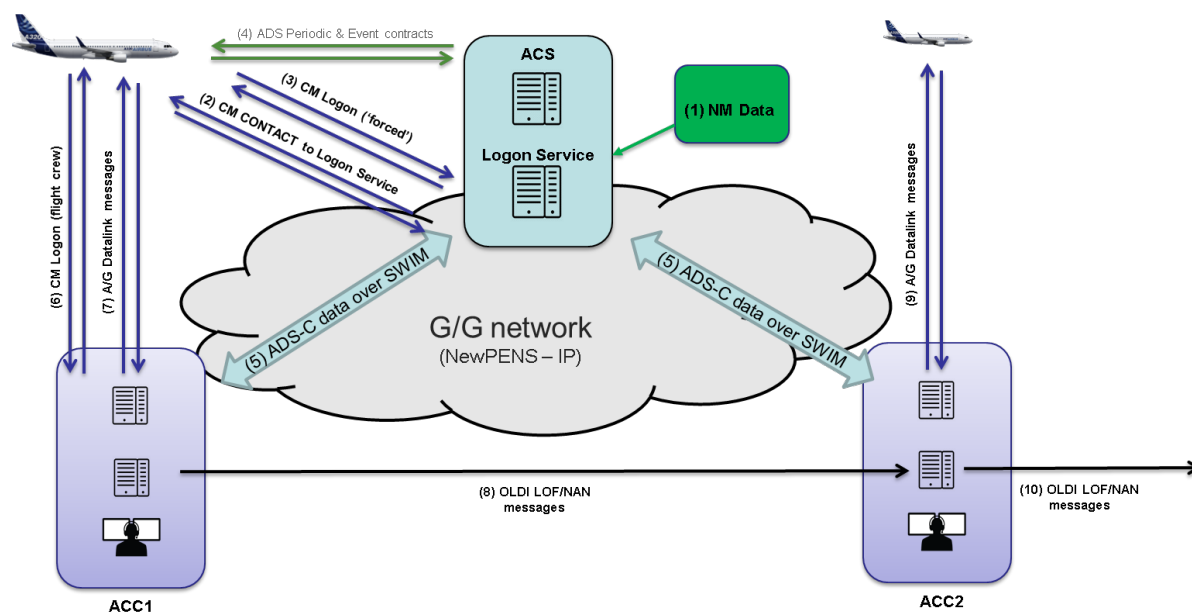


Figure 18: [Option 11] CM-logon Request to ACC1 and 'forced' CM-Logon to Logon Service

Pros	Cons	Impact Assessment
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1. Same flight deck logon procedures applied for B2 as for current B1 (no changes) 2. Non-dependency on manual flight deck CM-logon to perform ADS-C (Autologon) independent of crew actions 3. ADS-C/EPP data is available to all users after forced CM-logon to ACS 4. Logon Service can store all address information – a data broker or backup database of addresses	1. All ACCs need to support CM-logon with B1+B2 2. Deviates from ICAO/EUROCAE standards as CM-contact would be used without a CM-logon 3. Issues additional A/G CM-contact and CM-logon for every flight 4. Depends on quality of external data to Logon Service to trigger the CM-contact as early as possible	ANSP/DLFEP to review: 1. All centres need to support ATS B2 CM (CRO-1159, CRO-1212, CRO-1213, CRO-1214) ACS and Logon Service: 2. ACS is SWIM client of Logon Service 3. Needs to comply to CM-server requirements 4. Needs operational information to determine when to trigger the CM-contact for departing or inbound flights 5. CM-contact depends on the quality and availability of externally received addressing information 6. Define a means to retrieve stored CM logon address information Airspace Users: 1. Not supported by some current certified avionics 2. Unknown impact on certification requirements due to deviation from standards 3. Requires updates of avionics that reject CM-contact without a logon 4. No impact on airborne ATC database, the GFD and the NSAP address of the Logon Service would be indicated in the uplinked CM-contact Regulations and Standards: 1. ICAO/EUROCAE standards modification unfeasible for the CP1 industrial target date, proposal could be forwarded to WG78.
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A.2.12. [Option 12] CM-logon to ANSP (ACC1) and ACS ADS-C Contract without Logon

Description The flight deck crew initiates a CM-logon request containing B1 and B2 logon information which is sent to the first ACC on the flight trajectory. The ACS initiates the ADS-C contract with the aircraft without any CM-logon and relies on available addressing and available NM flight status and positional data. ACCs will receive EPP data from the ACS over the SWIM interface using NewPENS.

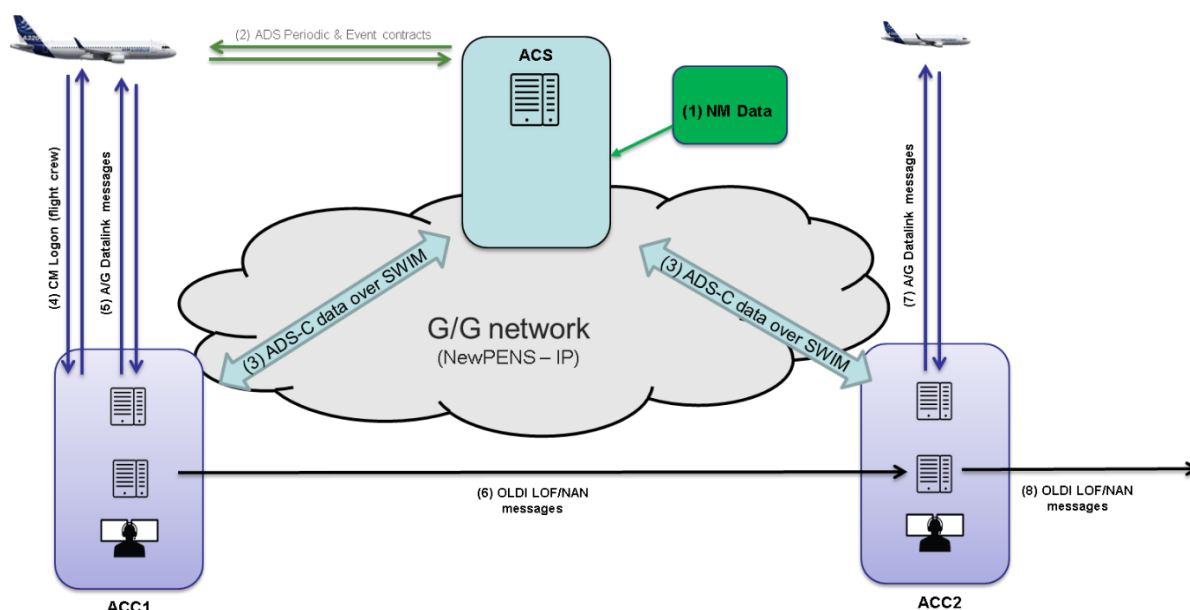


Figure 19: [Option 12] CM-logon Request to ACC1 and ADS-C Contracts without Logon

Pros	Cons	Impact Assessment
1. Same flight deck logon procedures applied for B2 as for current B1 (no changes) 2. Non-dependency on manual flight deck CM-logon to perform ADS-C 3. ADS-C/EPP data is available to all after ACS ADS-C contract establishment	1. All ACCs need to support CM-logon with B1+B2 2. Depends on quality of external data to ACS to trigger ADS-C as early as possible 3. Depends on the quality and availability of addressing information 4. Against ED228 DLIC-OR-6 5. No 'consent' from aircraft to downlink ADS-C data 6. Loss of CM services between aircraft and ACS (logon, applications, versions)	ANSP: 1. All centres need to support ATS B2 CM (CRO-1159, CRO-1212, CRO-1213, CRO-1214) ACS: 1. Implement A/C tracking function using NM data to start the ADS-C connection 2. Depends on quality of external data to ACS 3. Synchronization of addressing information needs to be developed Airspace Users: 1. Uncertain behaviour of all B2 avionics (Airbus/Honeywell do accept ADS-C contracts without logon) 2. No impact on current B1 fleet Regulations and Standards: 1. Against ED228 DLIC-OR-6

A.3. Evaluation of the Logon Options

A.3.1. Selection Criteria and Scoring Methodology

The below 20 selection criteria have been defined to evaluate the 12 options. Each option was scored against each criterion as follows: No, completely disagree (-2); No, partially disagree (-1); Neutral (0); Yes, partially agree (+1); Yes, fully agree (+2). As the first selection criteria is considered as a pre-requisite for the CONOPS, the only possible answers were 'Yes' or 'No'.

1. Does the option enable the flight crew to logon with one single manual CM Logon for all the Datalink Applications they intend to use? (1 logon procedure for all applications)
2. Does the option enable the flight crew to Logon with a single address for all the European airspace? (logon to 1 address)
3. Does the option avoid the flight crew to perform additional manual Logon per flight under non-nominal situations (e.g., failure/loss of CM Logon data, loss of LOF/NAN messages)? (1 logon procedure per flight)
4. Does the option enable the flight crew to make the manual CM Logon, at an early stage and at the time of their choice (AIPs to be adapted)?
5. Does the option apply the same CM Logon procedure for B1 and B2 aircraft (regardless of applications and versions supported by the avionics)?
6. Does the option support a scenario, where not all ANSPs and ATSUs deploy possible changes to CM Logon Management?
7. Does the option support a scenario where ANSPs (each individual ANSP) deploy changes to CM Logon Management independent of the operational usage of ADS-C functions?
8. Does the option accommodate Lower Airspace ATSUs to participate in ADS-C without running their own Datalink and CPDLC infrastructure?
9. Does the option encourage early ADS-C adopters and enable access to ADS-C data before a full European roll-out has been achieved?
10. Does the option ensure availability of full Logon information for downstream ATSUs, independent of Capabilities and DL Applications used by upstream ATSUs?
11. Does the option contribute to maximize ADS-C Connection rates?
12. Does the option limit major changes to ground systems (for the Logon management, in particular to core FDP functionality) – excluding SWIM changes? – this criterion only applies to the ANSPs interested on connecting/using the ACS
13. Does the option avoid changes to ground systems to those ANSPs not connecting to ACS? – this criterion only applies to those ANSPs not interesting on connecting/using the ACS
14. Does the option avoid major changes to avionics or re-certification? Candidate as pre-requisite – Ask to airframers about the auto-logon and airlines
15. Can the option deal with re-routing situations, which may change the initial DL ATSU during execution phase of the flight?
16. Can the option deal with different filters (i.e., Logon lists for ADS-C and/or CPDLC) for the ADS-C Common Service and CPDLC operations of individual ATSUs?
17. Can the option make Logon data available to Clients (e.g., ATSUs) on request, in addition to potential Logon Forwarding Mechanisms?
18. Does the option minimise the impact to existing standards and regulations?
19. Does the option avoid the need for further validation and/or demonstration? (compared to what has been done under SJU projects)
20. Does the option reduce the complexity of the ACS/CM server (service availability, SWAL)?

A.3.2. Scoring Outcome

The outcome of the scoring of those organisations that participated is presented in Figure 20.

	Option 1	Option 2	Option 3	Option 4	Option 5	Option 6	Option 7	Option 8	Option 9	Option 10	Option 11	Option 12
ANS CR	-7	-10	-10	-6	0	-4	-3	5	-6	3	11	8
AFR	5	2	1	3	13	4	5	1	17	4	3	1
Lufthansa	11	11	14	8	11	10	14	10	17	6	7	10
ENAV	0	-11	-8	0	-5	-10	-9	-4	1	9	5	1
DFS	-5	-12	-8	-3	11	3	5	17	24	8	12	7
MUAC	-1	-5	-1	0	16	9	18	20	19	5	-2	7
NATS	6	2	2	10	13	11	11	11	12	-2	5	9
DSNA	-1	-5	-7	2	15	8	12	19	23	6	-4	-1
ENAIRE	13	-3	-5	11	5	1	-4	7	5	9	9	8
TOTAL SCORE	21	-31	-22	25	79	32	49	86	112	48	46	50

Figure 20: Scoring Outcome

Discussions during the various OEP WST 12.2 meetings showed that the interpretation of the criteria were quite different from one organisation to the other, due to difference in specific local practices and/or history with CPDLC implementation. Thus, some options were not favoured by some organisations were strongly supported by others.

A first step was achieved by making use of the scoring results to identify a short-list of 5 options among the 12. This short-list included the 3 “highest scored” (options 5, 8 and 9), plus 2 options (1 and 11) at the explicit request of some stakeholders considering some specific interests.

A.3.3. Detailed Analysis

A.3.3.1. Review of Logon Options

Several meetings were organised to further review the short-listed options. OEP WST12.2 members concluded an extensive review of the pros and cons of each option and converged on a scenario meeting the operational requirements of section 1.4.

'[Option 5] CM-logon to Logon Service and CM-forward to ANSP (ACC1)' and '[Option 8] CM-logon to Logon Service and CM-contact to ANSP (ACC1)' (as well as options 6 and 7) require enhancing the Logon Service with FDP capability so it can target the right ACC at the right time with the appropriate logon information. These two options were considered overly complex especially considering the necessary measures for the Logon Service to avoid mis-targeting the ACC.

'[Option 1] CM-logon to ANSP (ACC1) and CM-forward to Logon Service' (all well as options 2, 3 and 4) makes the ACS (and all ADS-C users) dependent on the logon to first ACC and does not enable early access to ADS-C/EPP data. However, this provides the benefit of easily forwarding any received logon information to the Logon Service by using the CM-forward message which is already standardised by ICAO Doc 9880 [7].

'[Option 11] CM-logon to ANSP (ACC1) and Logon Service CM-contact' has the advantage of increasing logon rates and may be added to the Logon Service as described in section 6.4.3. However, this has to be further assessed, as it is not fully compliant with ICAO nor EUROCAE standards, which could outweigh the benefit.

'[Option 9] CM-logon to Logon Service and ANSP (ACC1) Request' is a CM-server ready to respond to ACC requests who best know when the data is required to establish. It implies a new SWIM request feature that has been formalised by workstream members and would require stakeholder buy-in.

A.3.3.2. AU Considerations

Depending on the selected option, flight crew procedures may need to change. In view of the described options, three key scenarios for AUs can be envisaged:

- AU Scenario 1: options requiring a flight deck manual logon to the first ACC. In this scenario, existing logon procedures as defined in the national AIPs apply.
- AU Scenario 2: options requiring a flight deck manual logon to the Logon Service instead of the first ACC. In this scenario the likely objective for airspace users will be that both B1 and B2 aircraft are handled equally, meaning that logon services for all aircraft should be provided by the Logon Service (AIPs shall be updated accordingly).
- AU Scenario 3: options requiring no flight deck actions for ADS-C ('autologon', 'forced' CM-logon, and ADS-C without logon) but for which flight crews lose control on data exchanges.

AU preference should be given to scenario 2 as in the long term, this envisages a single logon address simplifying the management of the airborne ATC centre databases. This implies that options 5 to 9 would be preferred. Moreover, an evolution of such as Logon Service should allow the flight crew to logon whilst on the ground (before take-off) or as soon as the aircraft is within ATN coverage. This would allow operators to insert the logon procedure in the routine flight crew Standard Operating Procedures / Flight Preparation, in comparison with the current random logon above FL250.

A.3.4. Combining the Best of all Options

The Logon Service is a combination of multiple options. It is a result of the scoring methodology, detailed assessments, and considerations for AUs. It enhances option 9 with features from options 1, 5 and 8 (and even 3) to address the “cons” that were identified for option 9, whilst maintaining the benefits of the others. Subsequent bilateral meetings were also held with ANSPs to involve additional experts that were not attending the OEP workstream meetings to also exchange views on this conclusion.

The combination involves an initial CM-logon to the Logon Service (step 1 in Figure 21). A timely SWIM request from the ACC (local trigger step 4 and request step 5) to receive the logon information indicating by which means the data should be returned (step 6a or 6b or 6c). The ACC request can be synchronous or asynchronous meaning that the Logon Service will respond immediately with any available information, or the Logon Service will respond once information becomes available within a set time.

To cover transitions, ACCs will need to maintain the capability to process direct CM-logon requests and forward logon information to the Logon Service (step 12), and also forward received logon information from OLDI LOF messages (step 9). Therefore, the Logon Service offers multiple means of communication with ACCs to best adapt to local needs and transitions (ground-ground use of CM-forward and/or SWIM, and even ground-air initiated CM-contact (step 6)).

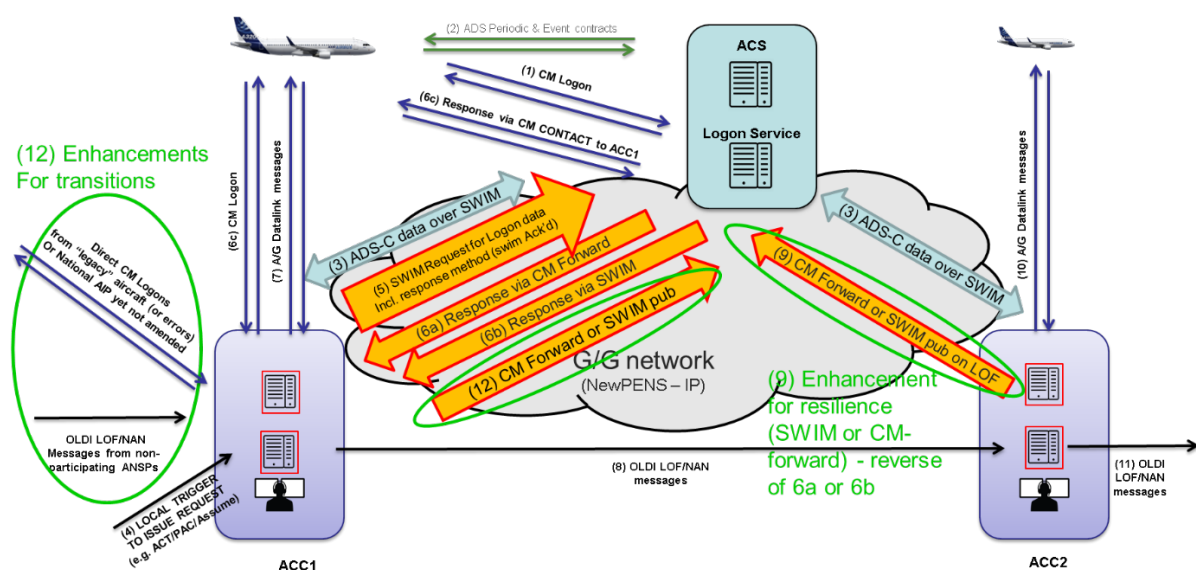


Figure 21: Combining Options for the Logon Service

Annex B. NDTECH and JCSP Scenarios Descriptions for OEP WST 12.2

The introduction of this CONOPS refers to NDTECH prioritisation of Scenario 2 acting a stepping-stone to Scenario 3. The below information recalls the content of those scenarios which were documented in 'NDTECH-22-6-8-Item 2.6 - ATS-B2 including ADS-C-EPP' and 'Action Paper - JCSP6/7 / AP01'.

Extract of NDTECH Paper (March 2022 – submitted by EUROCONTROL/SDM/S3JU):

The joint document presented three different scenarios for ATS-B2 development and deployment from a technical and operational perspective, including an analysis of missing technical standards. These scenarios are:

- Scenario 1 “No ATS B2”: this scenario corresponds to the non-activation of OEP 12.2 on ATS B2 (ATN B1 datalink improvements are performed by OEP 12.1).
- Scenario 2 “ADS-C/EPP development and deployment”: this scenario corresponds to the scope of CP1 AF6 combined with an ADS-C common service for ground distribution of EPP data.
- Scenario 3 “Extended ATS B2 development and deployment”: this scenario corresponds to the use of ADS-Cv1/v2 and CPDLCv2/v3 applications beyond the scope of CP1 and its timelines; it is also combined with an ADS-C common service for ground distribution of EPP data and includes the ATN/IPS datalink technology.

Scenario 1 NO ATS B2	Scenario 2 ADS-C/EPP DEPLOYMENT	Scenario 3 EXTENDED ATS B2 DEPLOYMENT
• JCSP does not engage in ATS B2 tasks	• CP1 Timeline • CPDLCv1 • ADS-Cv1 for EPP • ATN/OSI • Logon issues • B1/B2 compatibility • ADS-C Common Service	• CPDLC v2/v3 • ADS-C v1/v2 • Extend below FL285 • Extended operational use cases • ATN/IPS • Ground accommodation of mixed fleet • ... • ADS-C Common Service

Extract of JCSP Paper (December 2021 – prepared by EUROCONTROL/SDM/SJU):

Scenarios	Scenario 1: ATN B1 Improvements (no ATS B2)	Scenario 2: ADS-C/EPP Deployment (Part of ATS B2) CPDLC v1	Scenario 3: Extended ATS B2 Deployment (CPDLC v2/v3)
CP1 compliance	No	Yes (industrialisation target date in 2023 - full ground deployment and airborne forward-fit by 31/12/2027)	Compliant from the content perspective, but not by end 2027
Operational Drivers	- En-route - Operational improvements e.g., harmonised use of the messages sets	- En-route - Improved predictability and safety	- Integrated departure clearances, taxi and En-route (increased use of CPDLC) - Operational benefits from increased confidence in EPP data e.g., new ADS-C event contract types (PJ31) using Rev B standards - Improved vertical profiles for equipped aircraft thanks to full enhanced vertical clearances concept via CPDLCv2 (reduced fuel burn) - Integration of D-ATIS
Other Drivers	DLS IR mandatory use of CPDLCv1	CP1 AF6 Mandatory downlinking of EPP data as part of ATS B2 4DTRAD service (regardless of ADS-C version)	- Introduction of ATN/IPS in addition to ATN/OSI for global convergence and support of more aircraft types - Includes cybersecurity tools and procedures based on current validations - Make use of new ATN/IPS A/G communications services (SATCOM class A, LDACS, multilink)
Technical aspects	- Potential use of new VDL M2 frequencies - Potential offloading of AOC data from VDLM2 (aircraft multilink) to other means of communications e.g., SATCOM, terrestrial at airport - Resolution of current technical issues	- Deployment assumed to be on ATN/OSI only as ATN/IPS is under validation - Preferred deployment of ADS-C common service (not CP1 mandated) - CPDLC v1 - Ground accommodation of ADS-C/EPP - Potential introduction of new A/G technologies (i.e., SatCOM Class B)	- Transition from ATN/OSI towards ATN/IPS - Ground accommodation of mixed applications (different versions of CPDLC and ADS-C/EPP) and mixed ATN technologies (ATN/OSI and ATN/IPS)
Main Opportunities, Issues and Risks	Failure of European stakeholders to agree on harmonised deployment of ATS B2 applications in the short-term	- Compliant with CP1 in time - At present only one airframe manufacturer is committed to deploy ADS-C/EPP in time (over ATN/OSI)	- EUROCAE WG78 is progressing 'Rev B' of ED228 standard - Application of incentives - Consider the use of IM and DNRP

Scenarios	Scenario 1: ATN B1 Improvements (no ATS B2)	Scenario 2: ADS-C/EPP Deployment (Part of ATS B2) CPDLC v1	Scenario 3: Extended ATS B2 Deployment (CPDLC v2/v3)
		• The use of incentives is proposed to support timely deployment • Opportunity for improved vertical profiles thanks to TOD facilitation and initial enhanced vertical clearances via voice (fuel burn reduction) • Mixed support of ADS-C/EPP v1/v2 ⁷ • Some aircraft manufacturers will equip CPDLC v2 packed with EPP (which is not required by DLS IR). If this risk materialises, it means that either ANSPs accommodate CPDLC v2 logon or aircraft will not be capable to connect with the ground CPDLC⁸ • A/G datalink capacity needs to be reassessed based on deployment scenarios e.g., COVID impact, target fleet, offloading AOC from VHF, ADS-C common service	• Inclusion of D-ATIS together with DCL in order to rationalise the use of infrastructure • IPS is more flexible for future evolution: - Supports Multilink and planned by a SATCOM services provider - Includes security - Increases resilience • A/G datalink capacity needs to be reassessed based on deployment scenarios e.g., COVID impact, target fleet, offloading AOC from VHF, ADS-C common service

⁷ There are two existing versions of the same standard.

⁸ SDM is monitoring the situation on the ground (ANSPs) through the Monitoring View. It could be the case that ANSPs have already implemented a solution to accommodate CPDLC v2 connections.

Reference Material

1. OEP WST12.2 Topic Plan (Deliverable DEL4-1), Edition 1.0 dated 07 November 2022
2. SESAR PJ38 Project Internal Deliverables (ADS-C Common Service Requirements version 1.5 dated 7th June 2021, ADS-C Common Service SWIM ICD version 1.0 dated 27th October 2021)
3. EUROCAE ED-228A – Safety and Performance Requirements Standard for Baseline 2 ATS Data Communications (Baseline 2 SPR Standard) dated March 2016. This document is technically identical to RTCA DO-350A.
4. EUROCAE ED-229A – Interoperability Requirements Standard for Baseline 2 ATS Data Communications (Baseline 2 Interop Standard) dated March 2016. This document is technically identical to RTCA DO-351A.
5. EUROCAE ED-231A – Interoperability Requirements Standard for Baseline 2 ATS Data Communications and ATN Baseline 1 Accommodation (ATN Baseline 1 – Baseline 2 Interop Standard) dated March 2016
6. ED-120 – Safety and Performance Requirements Standard for Initial Air Traffic Data Link Services in Continental Airspace (SPR IC) dated May 2004
7. ICAO Doc 9880 – Manual on Detailed Technical Specifications for the Aeronautical Telecommunication Network (ATN) using ISO/OSI Standards and Protocols, 2nd Edition 2016
8. ICAO Doc 9896 – Manual for the Aeronautical Telecommunication Network (ATN) using Internet Protocol Suite (IPS), 2nd Edition 2015
9. EUROCONTROL Specification for On-Line Data Interchange (OLDI), Edition 5.0 dated 14th July 2020
10. NM Network 4DT CONOPS, Draft Edition 0.3 dated 28 September 2022
11. Commission Implementing Regulation (EU) 2021/116 of 1 February 2021 on the establishment of the Common Project One supporting the implementation of the European Air Traffic Management Master Plan
12. Commission Regulation (EC) No 29/2009 of 16 January 2009 laying down requirements on data link services for the single European sky (as amended)
13. SESAR Deployment Programme Edition 2022

Abbreviations

Additional abbreviations and acronyms used in this document are available in the EUROCONTROL Air Navigation Inter-site Acronym List (AIRIAL) which may be found online:

<http://www.eurocontrol.int/airial/definitionListInit.do?skipLogon=true&glossaryUid=AIRIAL>.

Term	Definition
A/C	Aircraft
ACC	Area Control Centre
ACDLS	ATS Common Datalink Service
ACS	ADS-C Common Service
ABI	Advance Boundary Information
ACH	ATC Flight Plan Change
ACS	ADS-C Common Service
ACT	Activation message (OLDI)
ANSP	Air Navigation Service Provider
ADEP	Aerodrome of Departure
ADES	Aerodrome of Destination
ADS-C	Automatic Dependent Surveillance – Contract
AEEC	Airlines Electronic Engineering Committee
AF6	ATM Functionality 6
AIP	Aeronautical Information Publication
AMQP	Advanced Message Queuing Protocol
ANSP	Air Navigation Service Provider
AOC	Airline Operational Communications
AoR	Area of Responsibility
ATC	Air Traffic Control

ATCO	Air Traffic Control Officer
ATN	Aeronautical Telecommunication(s) Network
ATS	Air Traffic Services
ATSU	Air Traffic Service Unit
AU	Airspace User
CBA	Cost Benefit Analysis
CONOPS	Concept of Operations
CP1	Common Project 1
CPDLC	Controller-Pilot Datalink Communications
CRO	Central Reporting Office
CWP	Controller Working Position
DLFEP	DataLink Front End Processor
ED	EUROCAE Document
EOBD	Estimated Off-Block Date
EOBT	Estimated Off-Block Time
EPP	Extended Projected Profile
FPL	Flight Plan
FL285	Flight Level 285
GAT	General Air Traffic
GFD	Ground Facility Designator
HTTPS	HyperText Transport Protocol Secured
ICAO	International Civil Aviation Organization
IMT	Information Management Team
IP	Internet Protocol
JCSP	Joint CNS Stakeholder Platform
LOF	Logon Forward (OLDI message)
MTCD	Medium-Term Conflict Detection

NAN	Next Authority Notified (OLDI message)
NDTECH	Network Directors of Technology Working Group
NOTAM	Notice to Airmen
NM	Network Manager
NSAP	Network Service Access Point
OEP	Operational Excellence Programme
OLDI	On-Line Data Interchange
RTCA	Radio Technical Commission for Aeronautics (US)
S3JU	SESAR Joint Undertaking
SDM	SESAR Deployment Manager
SES	Single European Sky
SESAR	Single European Sky ATM Research (Programme)
SWAL	Software Assurance Level
SWIM	System-Wide Information Management
ToC	Top of Climb
ToD	Top of Descent
TSAP	Transport Service Access Point
WG	Working Group
WST	Work Stream
XER	XML Encoding Rules

Table 2: Abbreviations table

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<http://www.eurocontrol.int/airial/definitionListInit.do?skipLogon=true&glossaryUid=AIRIAL>.



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